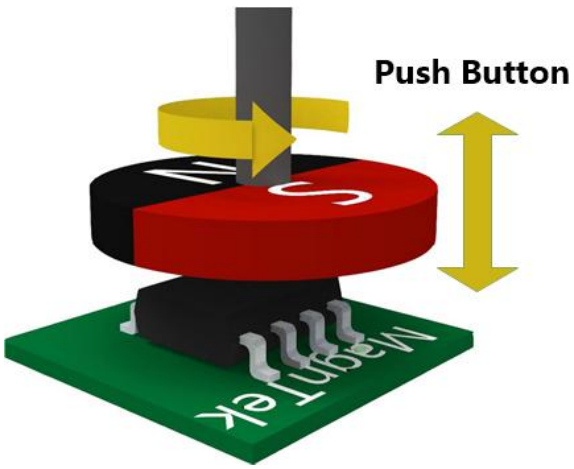
# MT6701

## Hall Based Angle Position Encoder Sensor

MagnTek | Now Part of NOVOSENSE

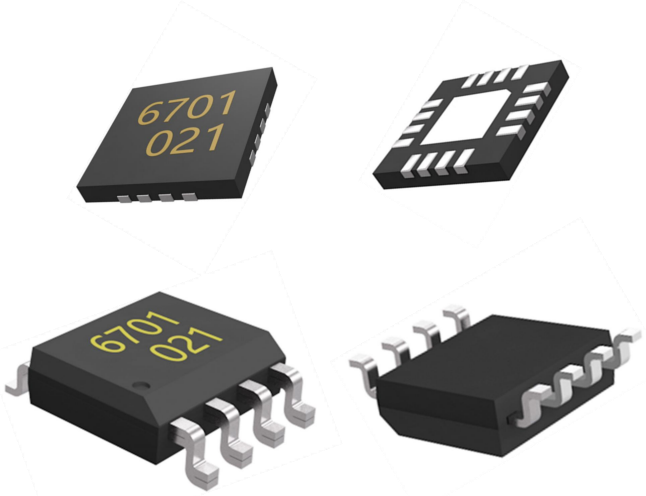
## Features and Benefits

- Based on Hall Sensing Technology with 0°~360° Full Range Angle Sensing
- 3.3~5.0V Operation Power Supply, build in EEPROM (Programing at VDD>4.5V)
- Independent Output Interface: I²C, SSI, ABZ, UVW, Analog/PWM and Push-Button Function on Z-Axis
- Maximum Rotation Speed 55,000 RPM
- Output Propagation Delay <5 us
- Incremental ABZ Resolution 1~1024 Pulses per Revolution (PPR) User Programmable
- UVW Output Resolution 1~16 Pole-Pairs per Revolution User Programmable
- RoHS Compliant 2011/65/EU
- SOP-8 and QFN3x3-16L Package




## Applications

- Absolute Linear Position Sensor
- Robotics Control
- Contactless Knob
- Contactless Potentiometer




## General Description

The MagnTek rotary position sensor MT6701 is an IC based on Hall sensing technology. The chip senses the Z-Axis component of the rotating magnet in the X-Y plane and delivers two sinusoidal output signals to the DSP which indicating the angle (α) between the sensor and the magnetic field direction.

The incremental ABZ output mode is available in this sensor series, making the chip suitable to replace various optical encoders. The maximum resolution is 1024 pulses/4096 steps per revolution in binary mode.

A standard I²C or SSI interface allows a host microcontroller to read the 14-bit absolute angle position data from MT6701. The absolute angle position is also provided as PWM output or linear analog signal proportional to VDD from a 12-bit DAC.

Additionally, the PUSH output indicates the fast air-gap changes between the MT6701 and magnet which can be used to implement a contactless pushbutton function in which the knob can be pressed to move the magnet toward the MT6701.

# Table of Contents

**Features and Benefits** ..... 1
**Applications** ..... 1
**General Description** ..... 1
**1 Pin Configuration** ..... 3
1.1 SOP-8 Package ..... 3
1.2 QFN-16 Package ..... 4
**2 Part Number List** ..... 5
**3 Function Diagram** ..... 6
**4 Absolute Maximum Ratings** ..... 6
**5 Electrical Characteristics** ..... 7
**6 Magnetic Input Specifications** ..... 9
**7 Output Mode** ..... 10
7.1 I/O Pin Configuration ..... 10
7.2 Reference Circuit for ABZ/UVW/PWM/Analog Output ..... 11
7.3 Quadrature A,B and Zero-Position Output Signal (ABZ Mode) ..... 12
7.4 UVW Output Mode ..... 17
7.5 Analog Output Mode ..... 18
7.6 Pulse Width Modulation (PWM) Output Mode ..... 19
7.7 I²C Interface ..... 20
7.7.1 I²C Timing Diagram ..... 21
7.7.2 I²C Read Angle Registers ..... 22
7.7.3 I²C Write Registers ..... 23
7.8 SSI Interface ..... 24
7.7.1 SSI Timing Diagram ..... 25
7.7.2 SSI Read Angle ..... 26
7.9 Pushbutton Output ..... 27
**8 Register Map & EEPROM Programming** ..... 28
8.1 Register Map ..... 28
8.2 EEPROM Programming ..... 32
**9 Mechanical Angle Direction** ..... 33
**10 Package Information** ..... 34
10.1 SOP-8 Package ..... 34
10.2 QFN-16 Package ..... 35
**11 Copy Rights and Disclaimer** ..... 36
**12 Revision History** ..... 37

# 1. Pin Configuration

## 1. 1 SOP-8 Package

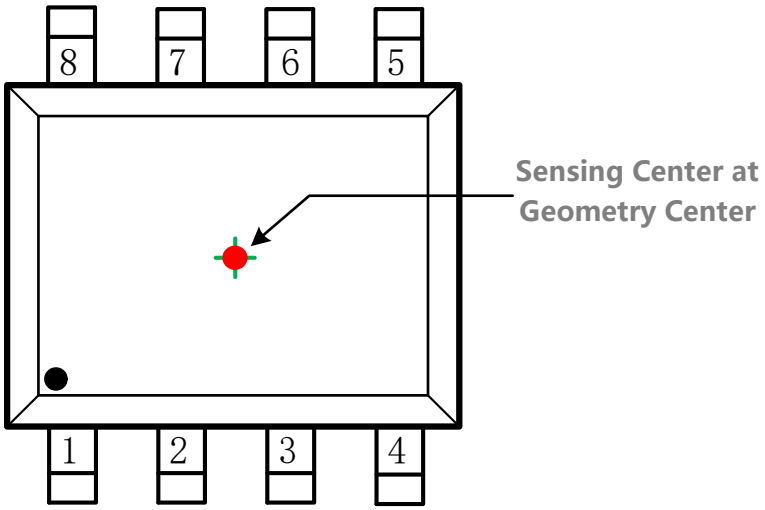



*Figure-1: Pin Configuration for SOP-8 Package*

**Pin List**

| Name | # | Type | Description |
|---|---|---|---|
| VDD | 1 | Power Supply | 3.3~5.0V Supply |
| MODE | 2 | Digital Input with Build in 200KΩ Pull-up Resistor | ABZ or I²C/SSI Selection |
| OUT | 3 | Analog/Digital Output | Analog or PWM Output |
| GND | 4 | Power Supply | Ground |
| PUSH | 5 | Digital Output | Push Button Function Output |
| A | 6 | Digital Input/output | Incremental Signal A/U or I²C Data 'SDA', or SSI Data 'DO' |
| B | 7 | Digital Input/output | Incremental Signal B/V or I²C Clock 'SCL', or SSI Clock 'CLK' |
| Z | 8 | Digital Input/output with Build in 200KΩ Pull-up Resistor | Incremental Signal Z/W or SSI 'CSN' |

***SOP-8 Reflow Sensitivity Classification: MSL-3***

## 1. 2 QFN-16 Package

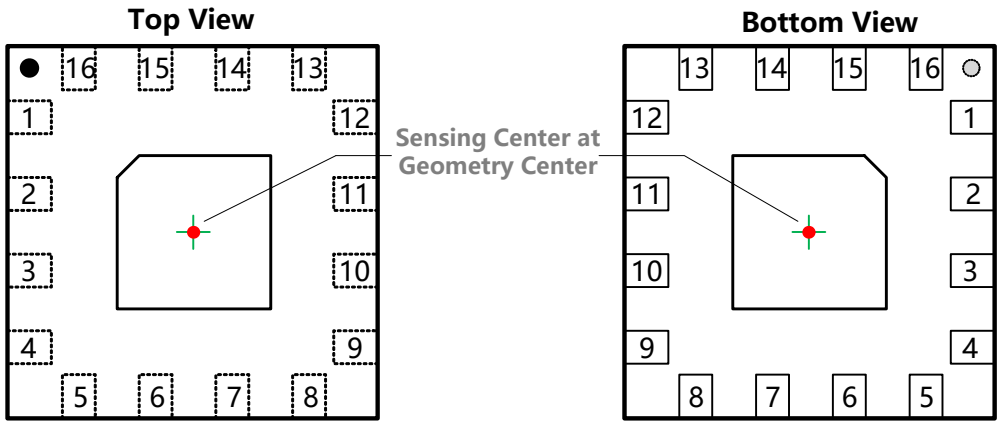



*Figure-2: Pin Configuration for QFN-16 Package*

**Pin List**

| Name | # | Type | Description |
|---|---|---|---|
| NC | 1 | - | Not Connected |
| NC | 2 | - | Not Connected |
| NC | 3 | - | Not Connected |
| NC | 4 | - | Not Connected |
| PUSH | 5 | Digital Output | Push Button Function Output |
| A | 6 | Digital Input/output | Incremental Signal A, or I²C data 'SDA', or SSI data 'DO' |
| B | 7 | Digital Input/output | Incremental Signal B, or I²C clock 'SCL', or SSI Clock 'CLK' |
| Z | 8 | Digital Input/output with Build in 200KΩ Pull-up Resistor | Incremental Signal Z, or SSI 'CSN' |
| W | 9 | Digital Output | Incremental Signal W or –Z |
| NC | 10 | - | Not Connected |
| U | 11 | Digital Output | Incremental Signal U or –A |
| V | 12 | Digital Output | Incremental Signal V or –B |
| VDD | 13 | Power Supply | 3.3~5.0V Supply |
| MODE | 14 | Digital Input with Build in 200KΩ Pull-up Resistor | ABZ or I²C/SSI Selection |
| OUT | 15 | Analog/Digital Output | Analog or PWM Output |
| GND | 16 | Power Supply | Ground |

***QFN-16 Reflow Sensitivity Classification: MSL-1***

## 2. Part Number List

MT6701CT is in an SOP-8 package (Reflow Sensitivity Classification: **MSL-3**), and both Tube Pack (**100pcs/Tube**) and Tape & Reel Pack (**3000pcs/Reel**) are provided. MT6701QT is in a QFN3x3-16L package (Reflow Sensitivity Classification: **MSL-1**), only Tape & Reel Pack (**1000pcs/Reel**) is provided.

**Part Number List**

| Part Number | Description |
|---|---|
| MT6701CT-STD | SOP-8 Standard Part Number: I²C, SSI; AB=1 PPR, Z=1 LSB;<br>Analog Output 0~360°; CCW Rotate Angle Increase; |
| MT6701QT-STD | QFN3x3 Standard Part Number: I²C, SSI; AB=1 PPR, Z=1 LSB; UVW=1 Pole-Pair<br>Analog Output 0~360°; CCW Rotate Angle Increase; |
| MT6701CT-AKD | AB=1,000 PPR, Z=4 LSB; Analog Output 0~360°; CCW Rotate Angle Increase; |
| MT6701CT-AKE | AB=1,000 PPR, Z=1 LSB; Analog Output 0~360°; CCW Rotate Angle Increase; |
| MT6701CT-ACD | AB=1,024 PPR, Z=4 LSB; Analog Output 0~360°; ; CCW Rotate Angle Increase; |
| MT6701CT-A200 | AB=200 PPR, Z=1 LSB; Analog Output 0~360°; CCW Rotate Angle Increase; |
| MT6701CT-A360 | AB=360 PPR, Z=1 LSB; Analog Output 0~360°; CCW Rotate Angle Increase; |
| MT6701CT-A400 | AB=400 PPR, Z=1 LSB; Analog Output 0~360°; CCW Rotate Angle Increase; |
| MT6701CT-A600 | AB=600 PPR, Z=4 LSB; Analog Output 0~360°; CCW Rotate Angle Increase; |
| MT6701CT-A800 | AB=800 PPR, Z=4 LSB; Analog Output 0~360°; CCW Rotate Angle Increase; |
| MT6701CT-STV | I²C, SSI; AB=1 PPR, Z=1 LSB; Analog Output 0~360°; **CW** Rotate Angle Increase; |
| MT6701QT-ACD | AB=1,024 PPR, Z=1 LSB; UVW=-A-B-Z;<br>Analog Output 0~360°; CCW Rotate Angle Increase; |

# 3. Functional Diagram

The MT6701 is manufactured in a CMOS standard process and uses advanced magnet sensing technology to sense the magnetic field distribution across the surface of the chip. The integrated magnetic sensing element array is placed around the center of the device and delivers a voltage representation of the magnetic field at the surface of the IC.

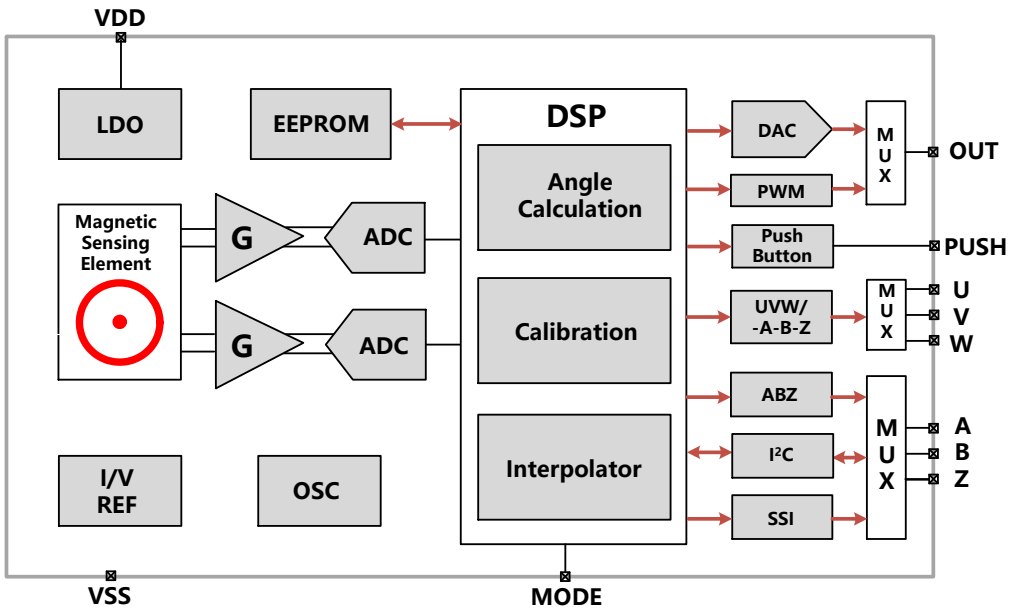



*Figure-3: Block Diagram*

Figure-3 shows a simplified block diagram of the chip, consisting of the magnetic sensing element modeled by two interleaved Wheatstone bridges to generate cosine and sine signals, gain stages, analog-to-digital converters (ADC) for signal conditioning, and a digital signal processing (DSP) unit for encoding. Other supporting blocks such as LDO, etc. are also included.

# 4. Absolute Maximum Ratings (Non-Operating)

Stresses beyond those listed under "Absolute Maximum Ratings" may cause permanent damage to the device. These are stress ratings only. Functional operation of the device at these or any other conditions beyond those indicated under "Operating Conditions" is not implied. Exposure to absolute maximum rating conditions for extended periods may affect device reliability.

| Name | Min. | Max. | Unit |
|---|---|---|---|
| DC Voltage at Pin VDD | -0.5 | 7 | V |
| Storage Temperature | -55 | 150 | ℃ |
| Operating Temperature | -40 | 125 | ℃ |
| Electrostatic Discharge (HBM) | - | ±6.0 | kV |
| Electrostatic Discharge (CDM) | - | ±1.5 | kV |

# 5. Electrical Characteristics

Operation conditions: Ta=-40 to 125℃, VDD=3.0~5.5V unless otherwise noted.

| Symbol | Parameter | Conditions/Notes | Min. | Typ. | Max. | Unit |
|---|---|---|---|---|---|---|
| VDD | Supply Voltage | - | 3.0 | 3.3~5.0 | 5.5 | V |
| Idd | Supply Current | - | | 10.0 | 14.0 | mA |
| LSB | Resolution (ABZ Mode) | N Steps per Cycle | | 360°/N | - | ° |
| INL | Integral Non-Linearity | **Note[1]** | - | ±1.0 | ±1.5 | ° |
| DNL | Differential Non-Linearity (ABZ Mode), **Figure-4** | | - | ±0.02 | - | ° |
| TN | Transition Noise (ABZ Mode) | 25℃ | - | 0.01 | | °rms |
| Hyst | Hysteresis (ABZ Mode) | | - | 0.088 | - | ° |
| $T_{PwrUp}$ | Power-Up Time | VDD Ramp<10us | - | - | 32 | ms |
| $T_{Delay}$ | Propagation Delay | Constant Speed | - | 5 | - | us |
| **Analog Output Specification** | | | | | | |
| $R_{OUT}$ | Analog Output Resistance | - | - | 15 | 30 | Ω |
| $R_L$ | Pull-Up or Pull-Down | - | 10 | - | - | KΩ |
| $C_L$ | Loading Capacitor | - | - | - | 1 | nF |
| $V_{Sat_High}$ | Saturation High Voltage | $I_{load}$=1mA | VDD-0.5 | - | - | V |
| $V_{Sat_Low}$ | Saturation Low Voltage | $I_{load}$=1mA | - | - | 0.5 | V |
| DAC_LSB | DAC LSB | 12-bit DAC | - | 0.0244 | - | %VDD |
| $V_{Noise}$ | Analog Output Noise | Ta=25℃, RMS Value excluding DAC Quantization Noise | - | - | 0.5 | mVrms |
| Erm | Ratiometric Error | **Note[2]** | -0.3 | - | 0.3 | % |
| **PWM Output Characteristics** | | | | | | |
| FPWM | PWM Frequency | Programmable | -5% @27℃ | 994.4 /497.2 | +5% @27℃ | Hz |
| $T_{Rise}$ | Rising Time | $C_L$=1nF | - | - | 1 | us |
| $T_{Fall}$ | Falling Time | $C_L$=1nF | - | - | 1 | us |

| Digital I/O Characteristics (Push-Pull Type in Normal Mode) | | | | | | |
|---|---|---|---|---|---|---|
| $V_{IH}$ | High Level Input Voltage | - | 0.7*VDD | - | - | V |
| $V_{IL}$ | Low Level Input Voltage | - | - | - | 0.3*VDD | V |
| $V_{OH}$ | GPIO Output High Level | Push-pull (Iout=2mA) | VDD-0.5 | - | - | V |
| $V_{OL}$ | GPIO Output Low Level | Push-pull (Iout=2mA) | - | - | 0.5 | V |
| $I_{LK}$ | Input Leakage Current | - | - | - | ±1 | uA |
| $T_{RISE}$ | Digital Output Rise Time | Push-Pull, $C_{Load}$=20pf | | | 40 | ns |
| $T_{FALL}$ | Digital Output Fall Time | Push-Pull, $C_{Load}$=20pf | | | 40 | ns |

**Note[1]**: The typical error value can be achieved at room temperature and with no off-axis misalignment error. The maximum error value can be achieved over operation temperature range, at maximum air gap and with worst-case off-axis misalignment error.

**Note[2]**: The analog output is by design ratiometric, i.e., it is proportional to the supply voltage VDD. The ratiometric error is calculated as follows.

$$\mathrm{Erm} = [\frac{Vout(V_{DD})}{V_{DD}} - \frac{Vout(5\mathrm{V})}{5V}] \cdot 100\%$$

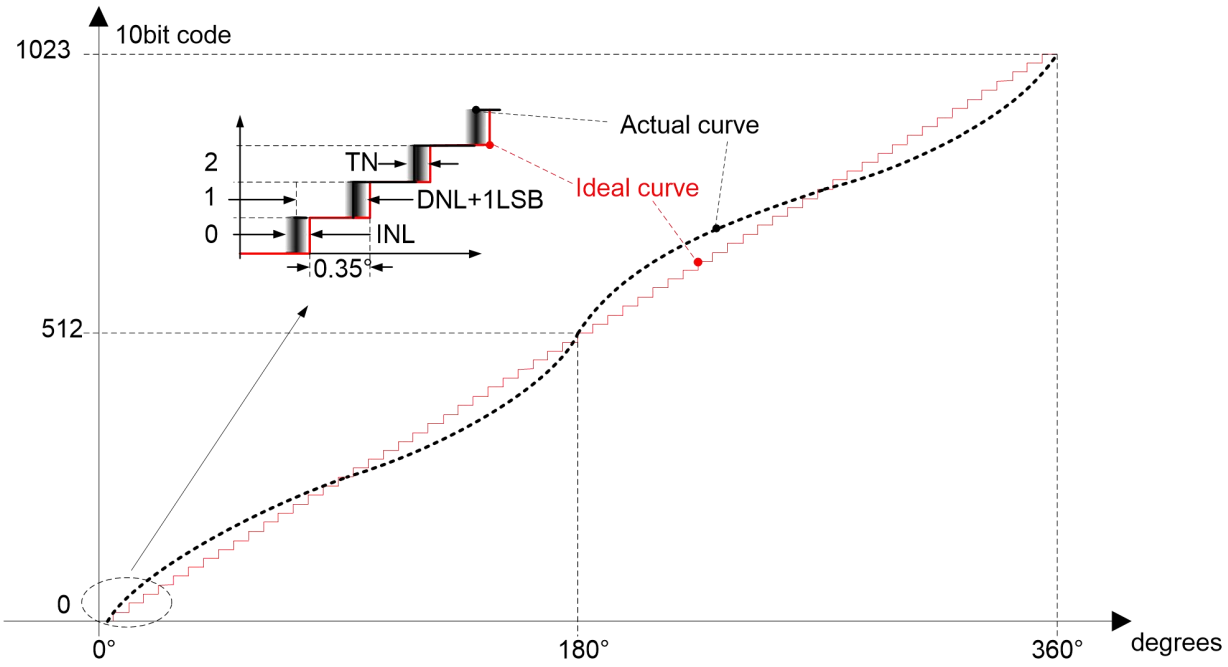



***Figure-4: Drawing Illustration INL, DNL and TN (for 10-bit case)***

# 6. Magnetic Input Specifications

Operation conditions: Ta=-40 to 125℃, VDD=3.0~5.5V unless otherwise noted, two-pole cylindrical diametrically magnetized source.

| Symbol | Parameter | Conditions/Notes | Min. | Typ. | Max. | Unit |
|---|---|---|---|---|---|---|
| Dmag | Diameter of Magnet | Recommended Magnet: Ø6mm x 2.5mm for Cylindrical Magnets | - | 6.0 | - | mm |
| Tmag | Thickness of Magnet | | - | 2.5 | - | mm |
| Bpk | Magnetic Input Field Amplitude Cross MT6701 | Measure at the IC Surface | 200 | - | 1,000 | Gauss |
| AG | Air Gap | Magnetic to IC Surface Distance | 0.5 | 1.0 | 2.0 | mm |
| RS | Rotation Speed | | - | - | 55,000 | RPM |
| DISP | Off Axis Misalignment | Misalignment Error Between Sensor Sensing Center and Magnet Axis (See **Figure-5**) | - | - | 0.3 | mm |
| TCmag1 | Recommended Magnet Material and Temperature Drift Coefficient | NdFeB (Neodymium Iron Boron) | - | -0.12 | - | %/℃ |
| TCmag2 | | SmCo (Samarium Cobalt) | - | -0.035 | - | |

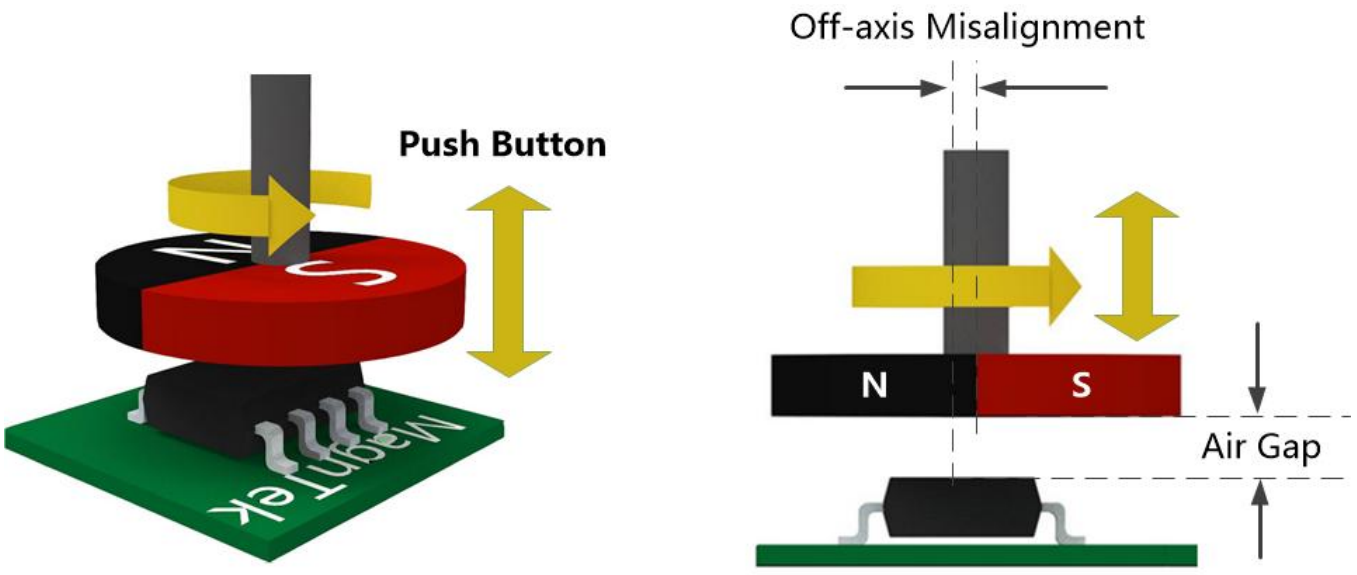



*Figure-5: Magnet Arrangement*

# 7. Output Mode

The MT6701 provides ABZ, UVW, Analog and PWM at output pins, also angle position data could be transferred by I²C or SSI interface. A PUSH output indicating pushbutton function is also provided.

## 7.1 I/O Pin Configuration

For SOP-8 package, ABZ/UVW (Single-end), I²C and SSI are configured to Pin.6, Pin.7 and Pin.8. Analog and PWM output is configured to Pin.3.

***SOP-8 Package I/O Pin Configuration***

| Pin# | I²C | SSI | ABZ | UVW | PWM | Analog | Push |
|---|---|---|---|---|---|---|---|
| 3 | | | | | PWM | Analog | |
| 5 | | | | | | | Push |
| 6 | SDA | DO | A | U | | | |
| 7 | SCL | CLK | B | V | | | |
| 8 | | CSN | Z | W | | | |

For QFN-16 package, ABZ (Single-end), I²C and SSI are configured to Pin.6, Pin.7 and Pin.8. UVW and –A-B-Z are configured to Pin.11, Pin.12 and Pin.9. Analog and PWM output is configured to Pin.15.

***QFN-16 Package I/O Pin Configuration***

| Pin# | I²C | SSI | ABZ | ABZ+UVW | ABZ Differential | PWM | Analog | Push |
|---|---|---|---|---|---|---|---|---|
| 5 | | | | | | | | Push |
| 6 | SDA | DO | A | A | A | | | |
| 7 | SCL | CLK | B | B | B | | | |
| 8 | | CSN | Z | Z | Z | | | |
| 9 | | | | W | -Z | | | |
| 11 | | | | U | -A | | | |
| 12 | | | | V | -B | | | |
| 15 | | | | | | PWM | Analog | |

## 7.2 Reference Circuit for ABZ/UVW/PWM/Analog/Push Output

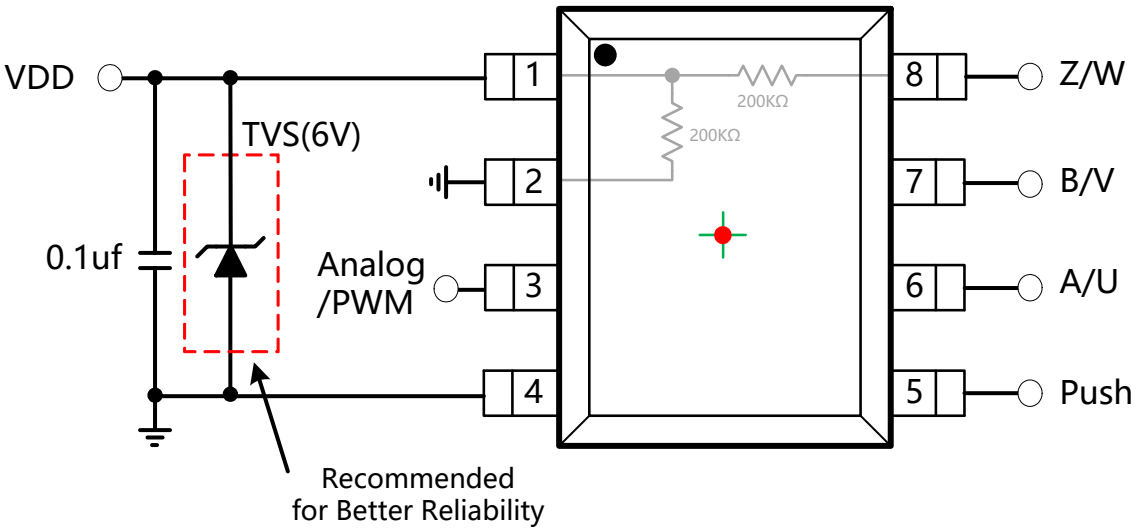



*Figure-6: MT6701CT(SOP-8) Reference Circuit for ABZ/UVW/PWM/Analog/Push Output*

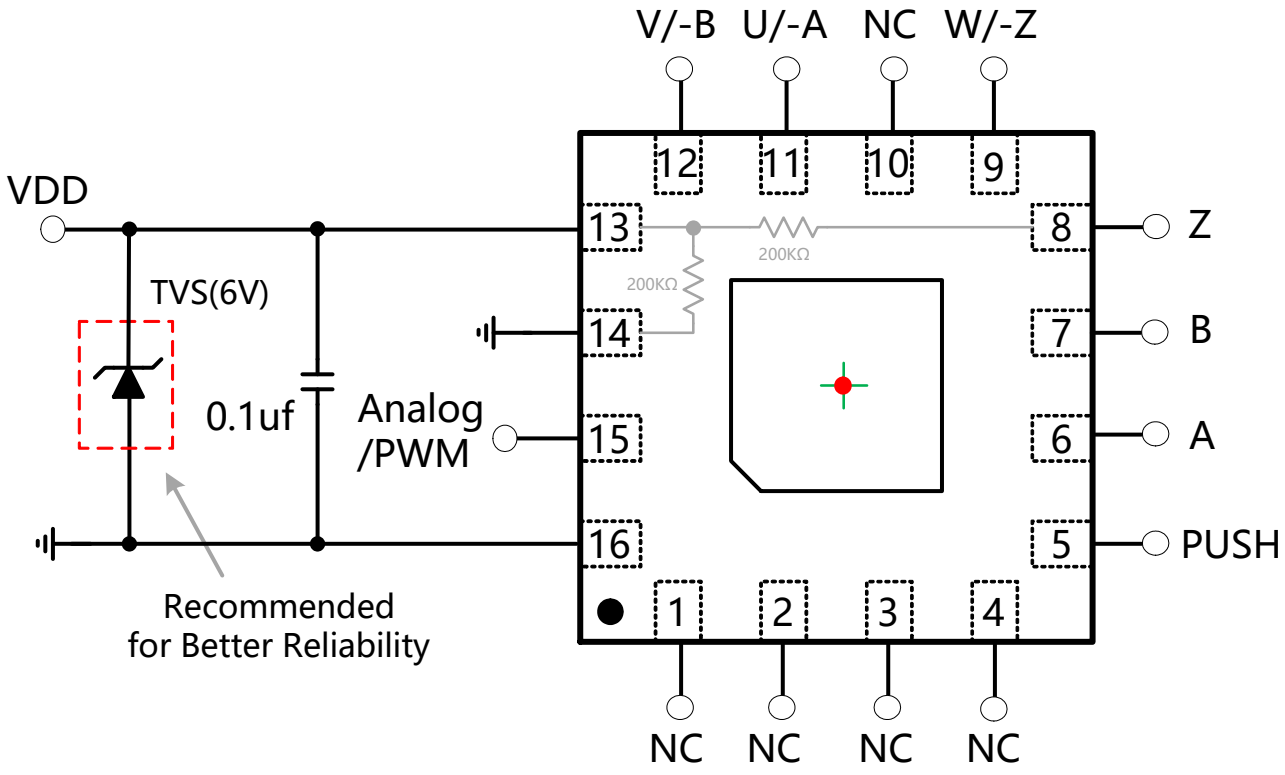



*Figure-7: MT6701QT(QFN-16) Reference Circuit for ABZ/UVW/PWM/Analog/Push Output*

## 7.3 Quadrature A, B and Index Output

As shown in Figure-8(1), when the magnet rotates counter-clock-wise (CCW), output B leads output A by 1/4 cycle, when the magnet rotates clock-wise (CW), output A leads output B by 1/4 cycle (or 1 LSB). Output Z indicates the zero position of the magnet.

After chip power-on, the ABZ output is blocked for 50ms to guarantee proper output.

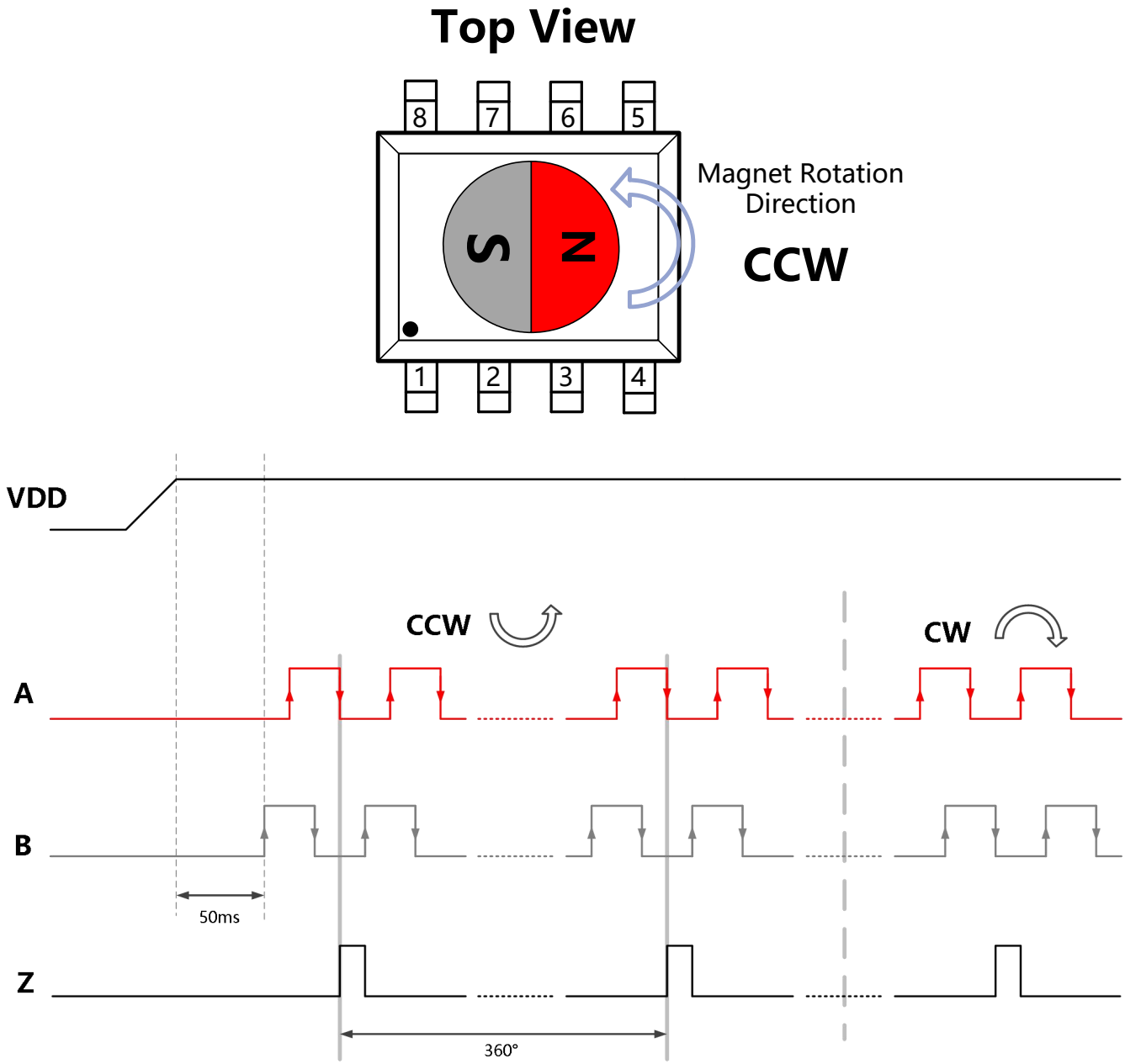



*Figure-8(1): ABZ Output Power On without Absolute Position Indicate Pulses*

After startup, there is an option of MT6701 to transmit a number of AB pulses which correspond to the actual absolute angle value as shown in Figure-8(2). Thus, the microcontroller gets the information about the absolute position.

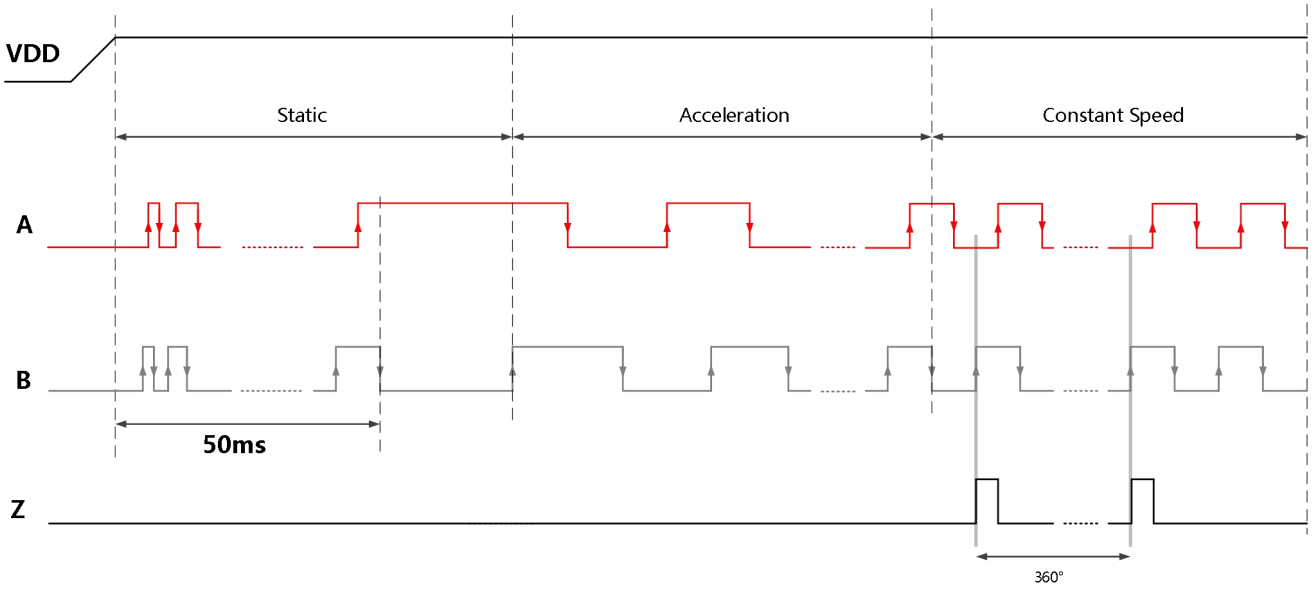



*Figure-8(2): ABZ Output Power On with Absolute Position Indicate Pulses*

Output Z indicates the zero position of the magnet, and it is user programmable. The pulse width of Z is selectable as 1, 2, 4, 8, 12, 16 LSBs and 180° as shown in Figure-9 and Figure-10. It is guaranteed that one Z pulse is generated for every rotation round.

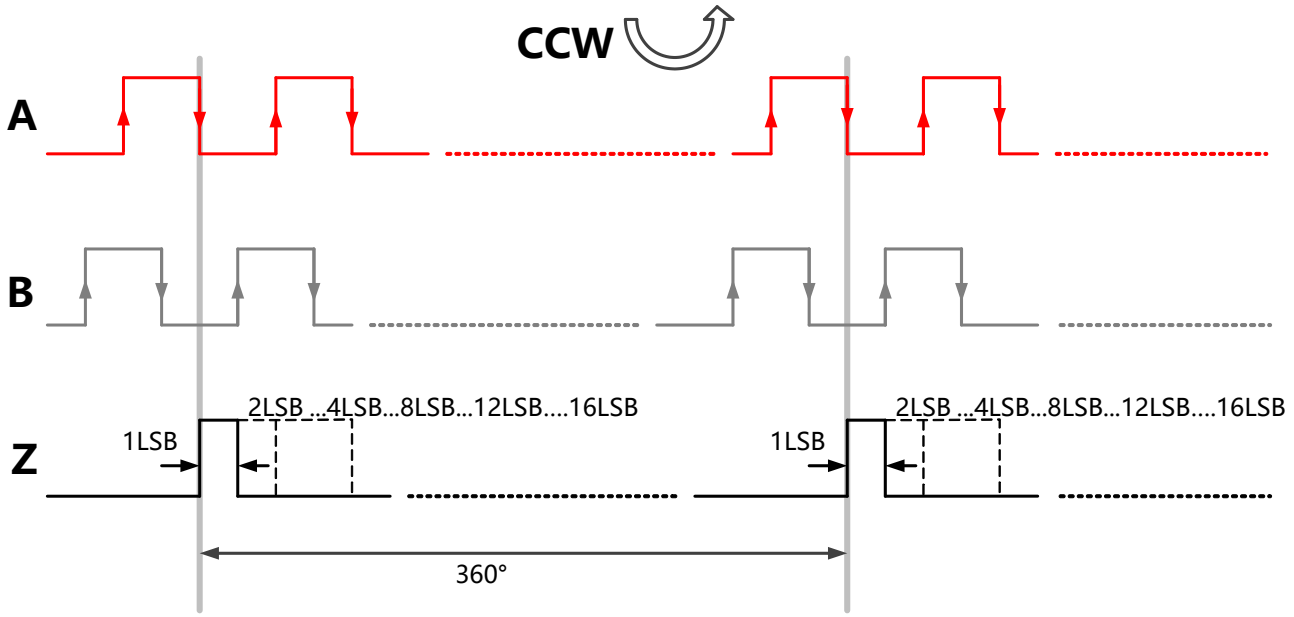



*Figure-9: Typical ABZ Output w/i Z Pules Width=1,2,4,8,12 and 16 LSBs*

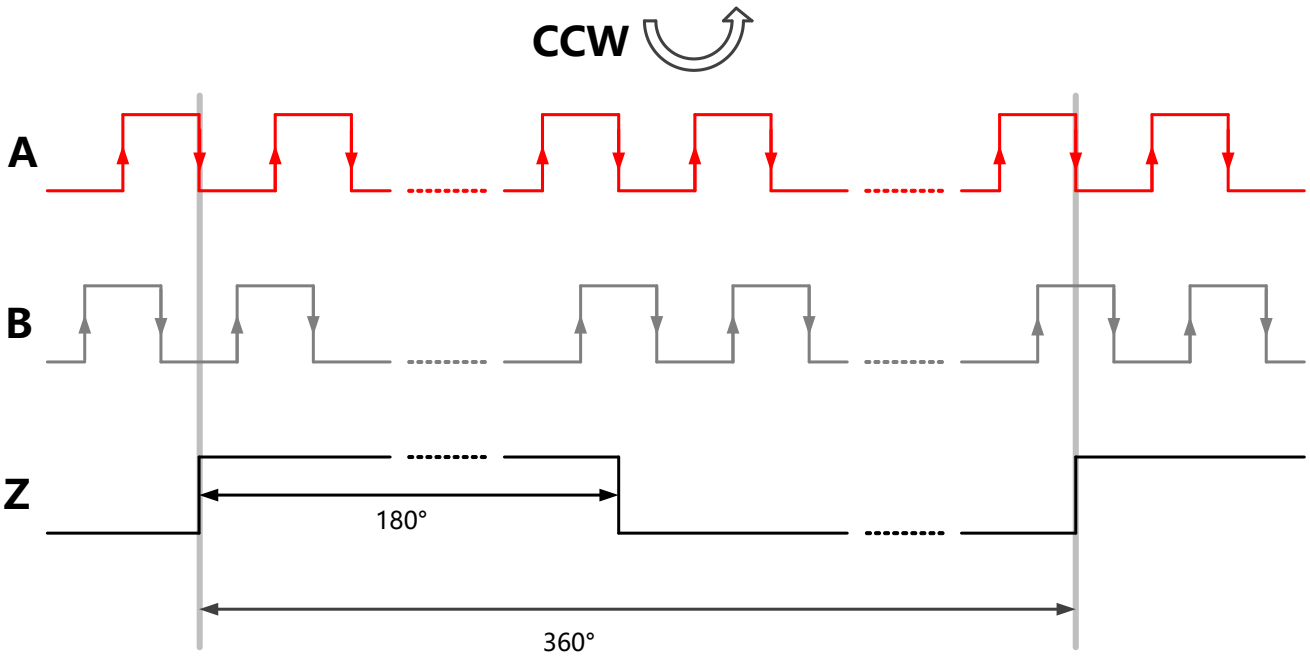



*Figure-10: Typical ABZ Output w/i Z pules width=180°*

ABZ resolution is user programmable as 1~1024 PPR any resolution. The relationship between binary bits, LSBs and PPR resolution of ABZ output are shown in Figure 11 and Figure 12.

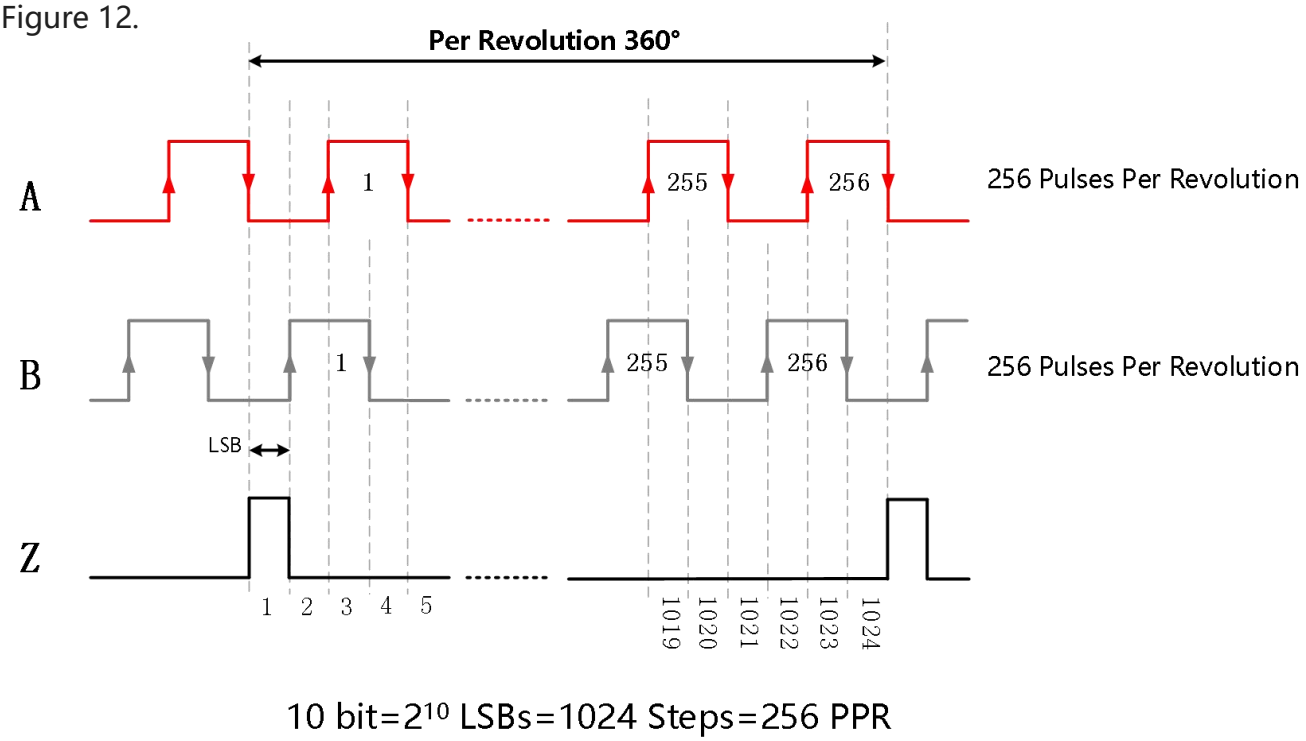



*Figure-11: ABZ Output Resolution=10 bit*

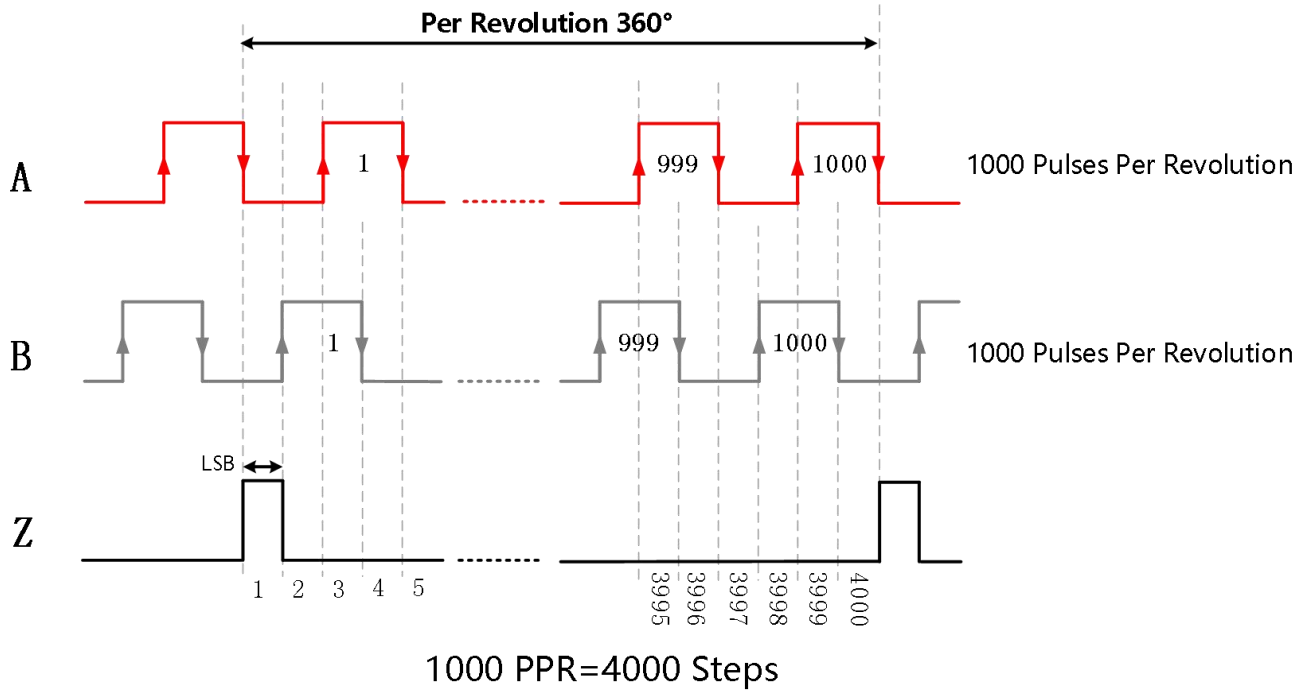



*Figure-12: ABZ Output Resolution=1000 PPR*

The Z/Index pulse width could be programmed

***Z/Index Pulse Width Register (EEPROM)***

| Reg. Z_Pulse_Width[2:0] | Width (LSBs) | Reg. Z_Pulse_Width[2:0] | Width (LSBs) |
|---|---|---|---|
| **000** | 1 | **100** | 12 |
| **001** | 2 | **101** | 16 |
| **010** | 4 | **110** | 180° |
| **011** | 8 | **111** | 1 |

The mechanical zero position could be programmed, it is a 12 bits data for 0~360°.

***Zero Position Register (EEPROM)***

| Register | bit7 | bit6 | bit5 | bit4 | bit3 | bit2 | bit1 | bit0 |
|---|---|---|---|---|---|---|---|---|
| **Zero_MSB** | NA | Z_PULSE_WIDTH[2:0] | | | ZERO[11:8] | | | |
| **Zero_LSB** | ZERO[7:0] | | | | | | | |

The resolution of ABZ could be programmed by a 10-bit register 'ABZ_RES'

***ABZ Resolution Register (EEPROM)***

| Register | bit7 | bit6 | bit5 | bit4 | bit3 | bit2 | bit1 | bit0 |
|---|---|---|---|---|---|---|---|---|
| **ABZ_RES** | UVW_RES[3:0] | | | | NA | NA | ABZ_RES[9:8] | |
| **ABZ_RES** | ABZ_RES[7:0] | | | | | | | |

**** To program EEPROM, please refer Chapter-8***

## 7.4 UVW Output Mode

The MT6701 provides U, V and W pulses which are 120° (electrical) out of phase as shown in Figure-13. The cycles of UVW per rotation can be programmed.

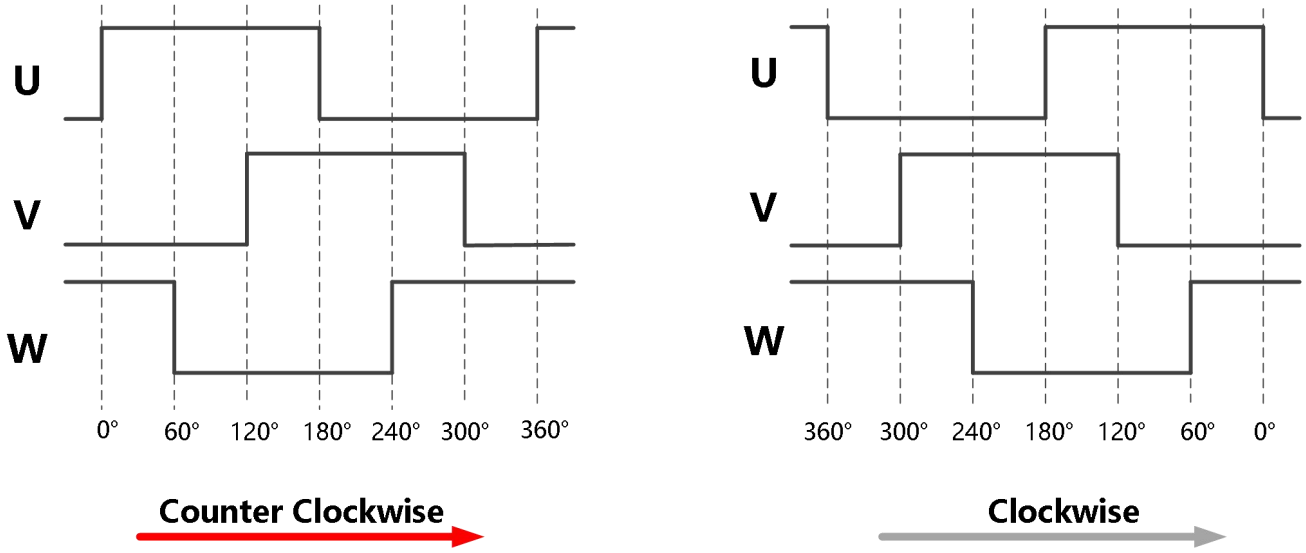



*Figure-13: Typical Output Waveform for UVW Mode*

***UVW Pole Pairs Register (EEPROM)***

| Reg. UVW_RES[3:0] | UVW Pole Pairs |
|---|---|
| 0000 | 1 |
| 0001 | 2 |
| 0010 | 3 |
| 0011 | 4 |
| 0100 | 5 |
| 0101 | 6 |
| 0110 | 7 |
| 0111 | 8 |
| 1000 | 9 |
| 1001 | 10 |
| 1010 | 11 |
| 1011 | 12 |
| 1100 | 13 |
| 1101 | 14 |
| 1110 | 15 |
| 1111 | 16 |

## 7.5 Analog Output Mode

The MT6701 provides a rail-to-rail linear analog output by a build-in 12-bit DAC as shown in Figure-14. It's a linear transfer function of absolute angle and output voltage. To enable analog output, register 'Output Mode' should be programmed to logic 'Low'.

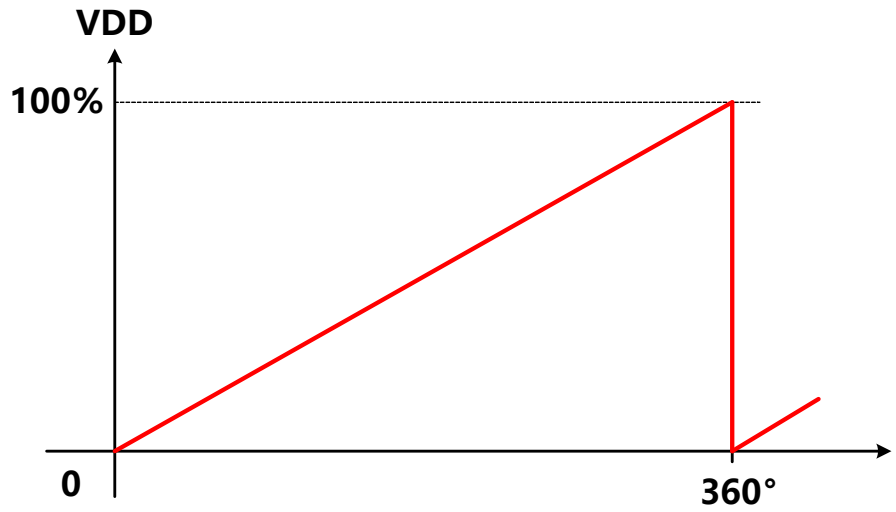



*Figure-14: Default Analog Output*

***Analog or PWM Output Control Register (EEPROM)***

| Reg. Output Mode | Pin.3 (SOP-8), Pin.15 (QFN-16) |
|---|---|
| 0 | Analog |
| 1 | PWM |

The angle and voltage value of start-point, Clamp_Low (0% or 10%) and Clamp_High (100% or 90%) could be user programmed, also the Zero Point could be user programmed as shown in Figure 15.

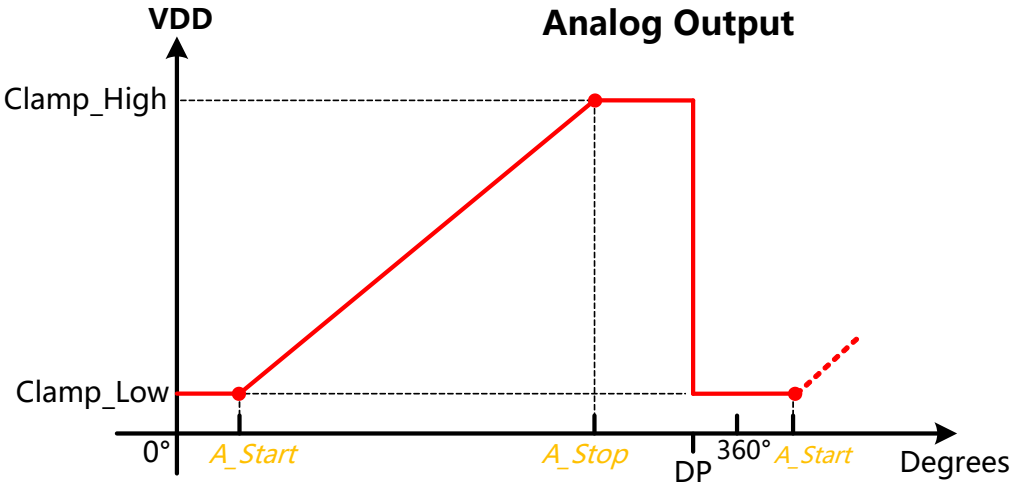



*Figure-15: Analog Output Transfer Function*

## 7.6 Pulse Width Modulation (PWM) Output Mode

The MT6701 provides a digital Pulse Width Modulation (PWM) output, whose duty cycle is proportional to the measured angle as shown in Figure-16.

The PWM output consists of a frame of 4119 PWM clock periods. The angle data is represented with 12-bit resolution in the frame. One PWM clock period represents 0.088° and has a typical duration of 244ns which also could be programmed to be 488ns.

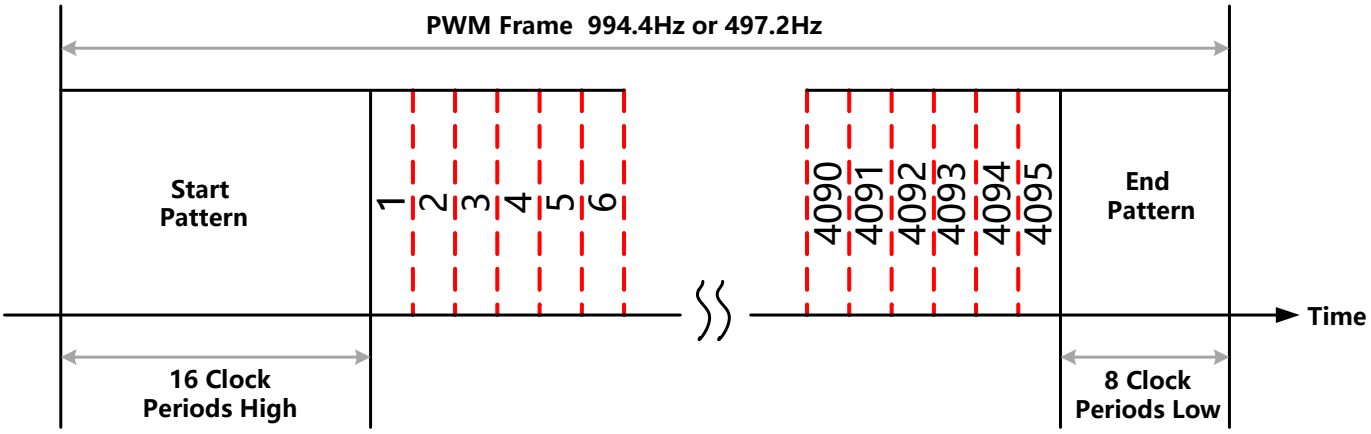



*Figure-16: PWM Output Frame*

***PWM Frequency (EEPROM)***

| Reg. PWM_FREQ | PWM Frame Frequency |
|---|---|
| **0** | 994.4 Hz |
| **1** | 497.2 Hz |

## 7.7 I²C Interface

The MT6701 provides a slave I²C interface for host MCU to read back digital absolute angle information from its internal registers. The reference circuit for I²C interface is shown in Figure-17 and Figure-18, whether the need for pull-up resistor on SCL is determined by MCU, for MT6701 SCL is a digital input.

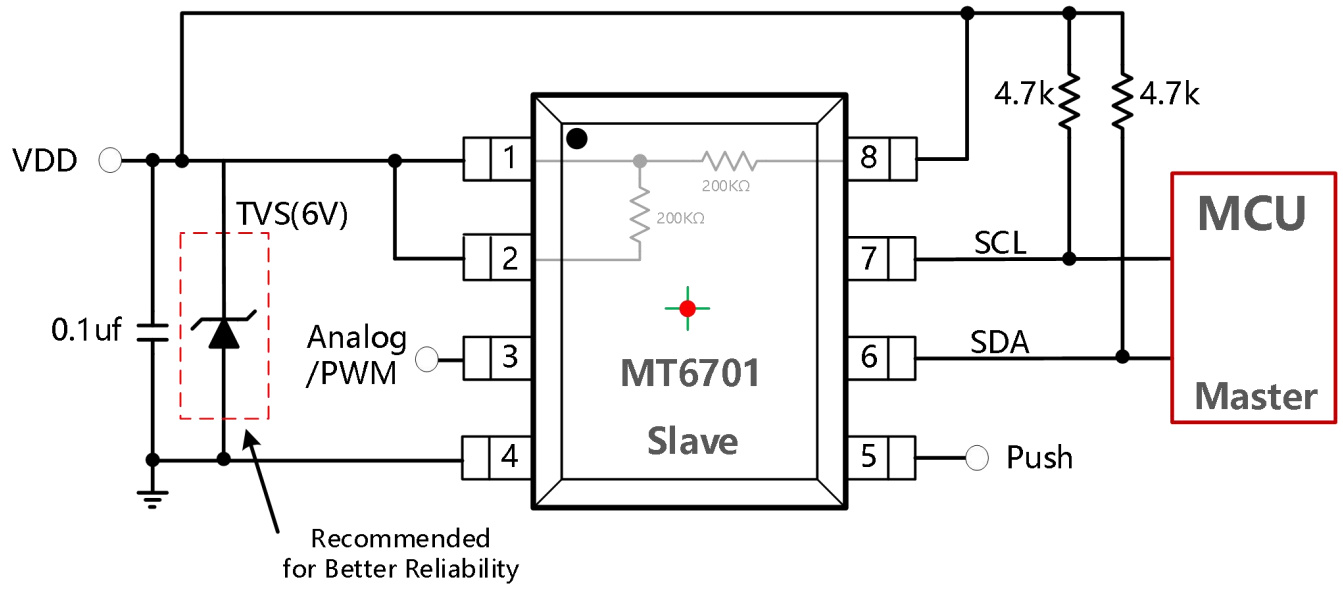



*Figure-17: I²C Reference Circuit of SOP-8 Package*

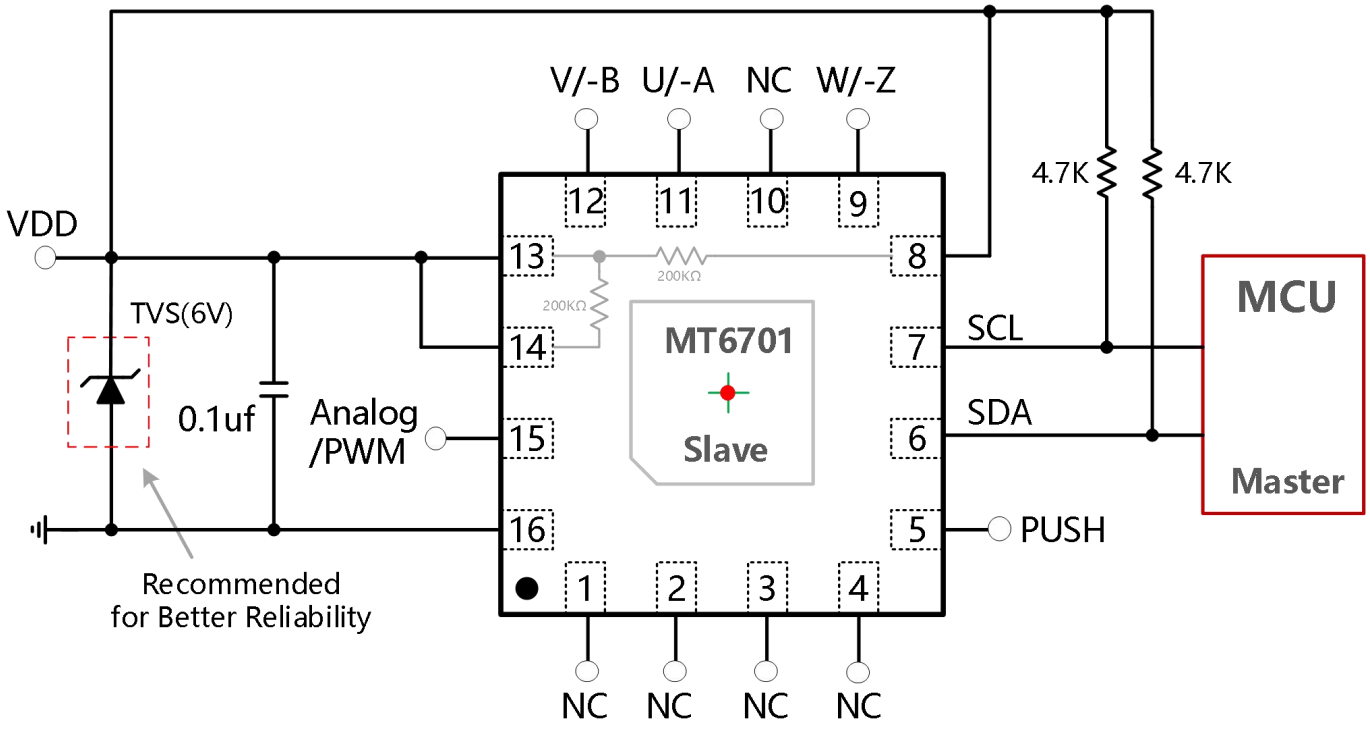



*Figure-18: I²C Reference Circuit of QFN3x3 Package*

### 7.7.1 I²C Timing Diagram

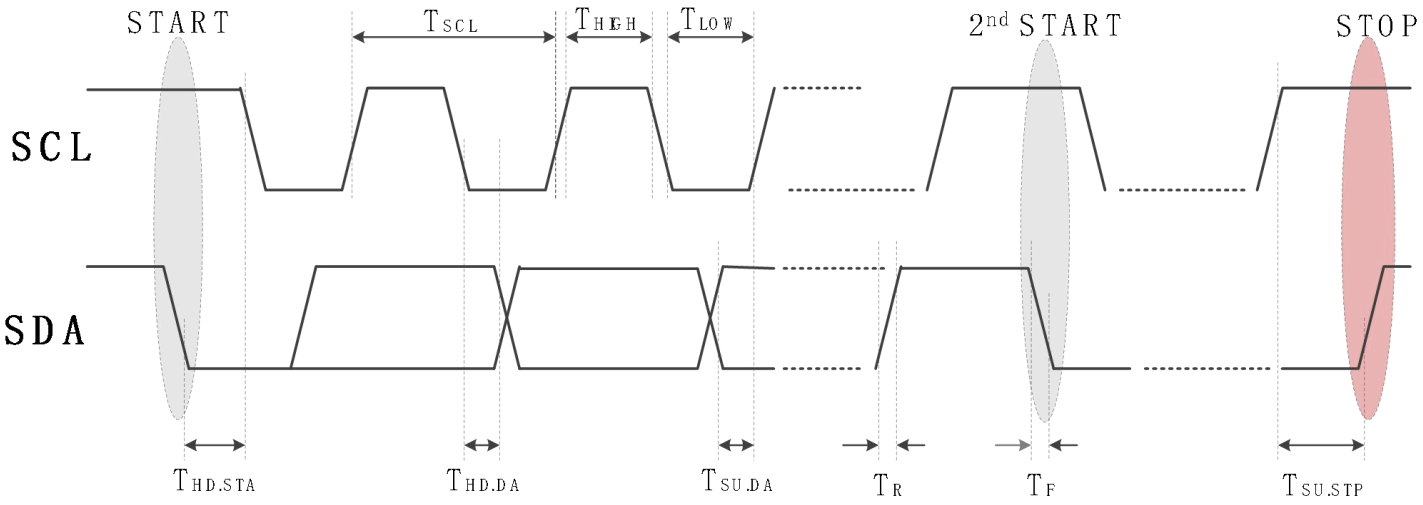



*Figure-19: I²C Timing Diagram*

*I²C Timing Parameter*

| Parameter | Notes | Min. | Max. | Unit |
|---|---|---|---|---|
| $T_{SCL}$ | SCL Clock Period | 1 | - | μs |
| $T_{HD.STA}$ | Hold Time of 'START' | 250 | - | ns |
| $T_{LOW}$ | Low Phase of SCL | 250 | - | ns |
| $T_{HIGH}$ | High Phase of SCL | 250 | - | ns |
| $T_{SU.DA}$ | Setup Time of SDA | 100 | - | ns |
| $T_{HD.DA}$ | Hold Time of SDA | 50 | - | ns |
| $T_R$ | Rising Time of SDA/SCL | - | 150 | ns |
| $T_F$ | Falling Time of SDA/SCL | - | 150 | ns |
| $T_{SU.STP}$ | Setup Time of 'Stop' | 250 | - | ns |

### 7.7.2 I²C Read Angle Registers

The default slave ID of MT6701 is b'0000110 in 7-bit binary form(It could be programmed to b'1000110) . The 14 bits angle data is stored in internal register 0x03 and 0x04. Please follow the I²C timing of Figure-20 to read the angle data from 0x03 and 0x04 registers.

***Note: Please read Register 0x03 first and then read 0x04***

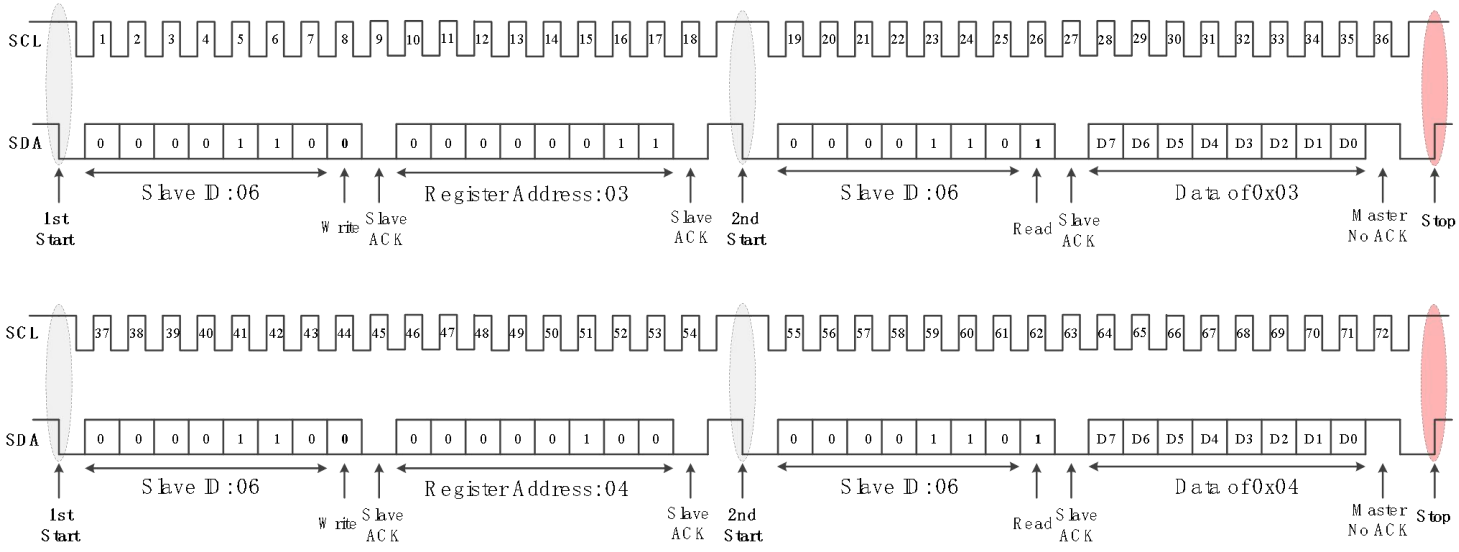



***Figure-20: I²C Single Byte Read***

***Angle Data Register***

| Reg. Addresss | Bit7 | Bit6 | Bit5 | Bit4 | Bit3 | Bit2 | Bit1 | Bit0 |
|---|---|---|---|---|---|---|---|---|
| **0x03** | Angle[13:6] | | | | | | | |
| **0x04** | Angle[5:0] | | | | | | NA | NA |

0~360° absolute angle θ could be calculated by the below formula:

$$\theta = \frac{\sum_{i=0}^{13} Angle<i> \bullet 2^{i}}{16384} \bullet 360°$$

### 7.7.3 I²C Write Registers

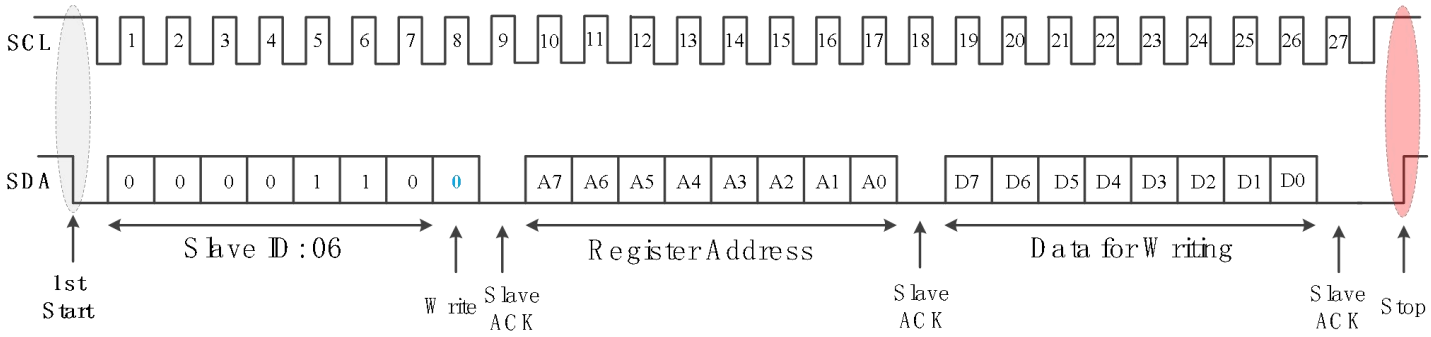



*Figure-21: I²C Write*

## 7.8 SSI Interface

The MT6701 also provides an SSI interface for host MCU to read back digital absolute angle information. The reference circuits for SSI interface are shown in Figure-22 and Figure-23.

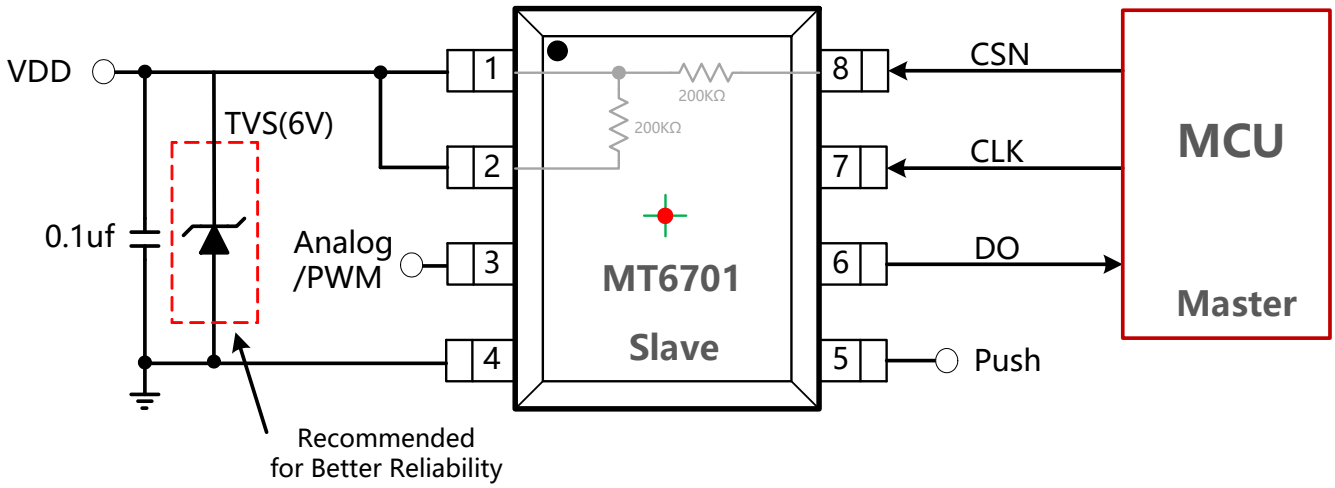



*Figure-22: SSI Interface Reference Circuit of SOP-8 Package*

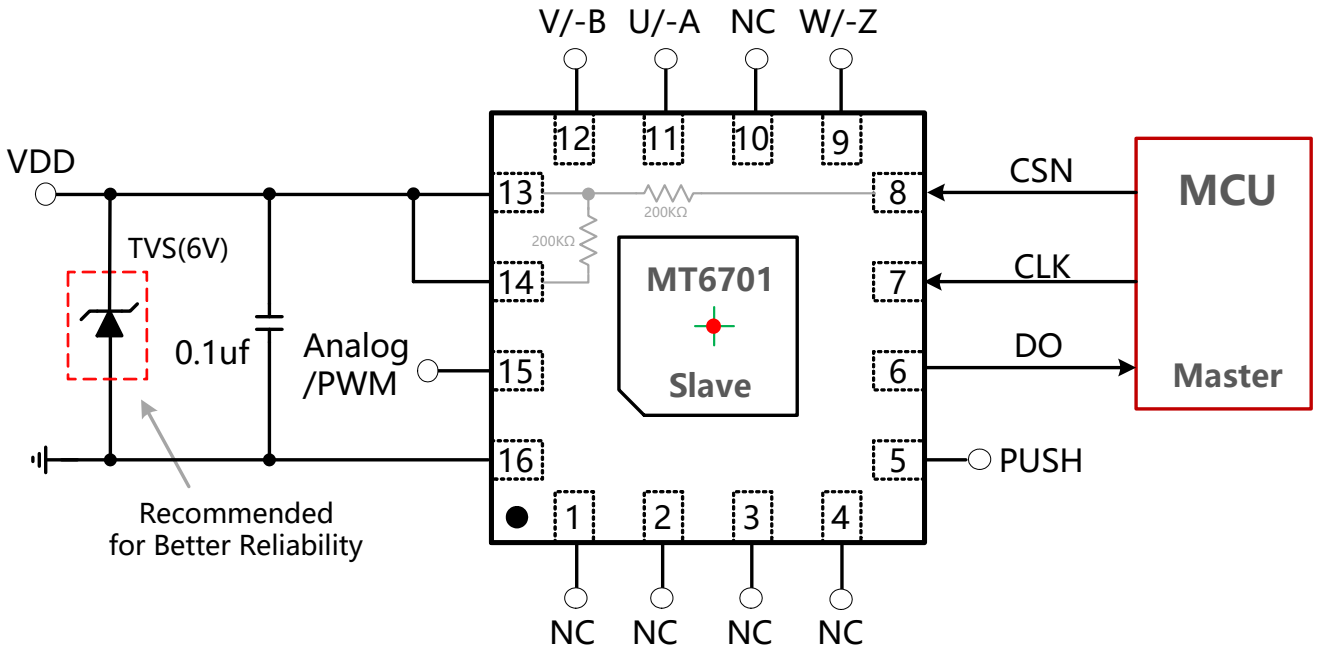



*Figure-23: SSI Interface Reference Circuit of QFN-16 Package*

### 7.8.1 SSI Timing Diagram

The MT6701 SSI is shown in Figure-24, a data transfer starts when CSN is pulled to logic 'Low'. The MT6701 transfers data on the rising edge of CLK, and the data transfer finally stops when CSN is pulled to logic 'High'

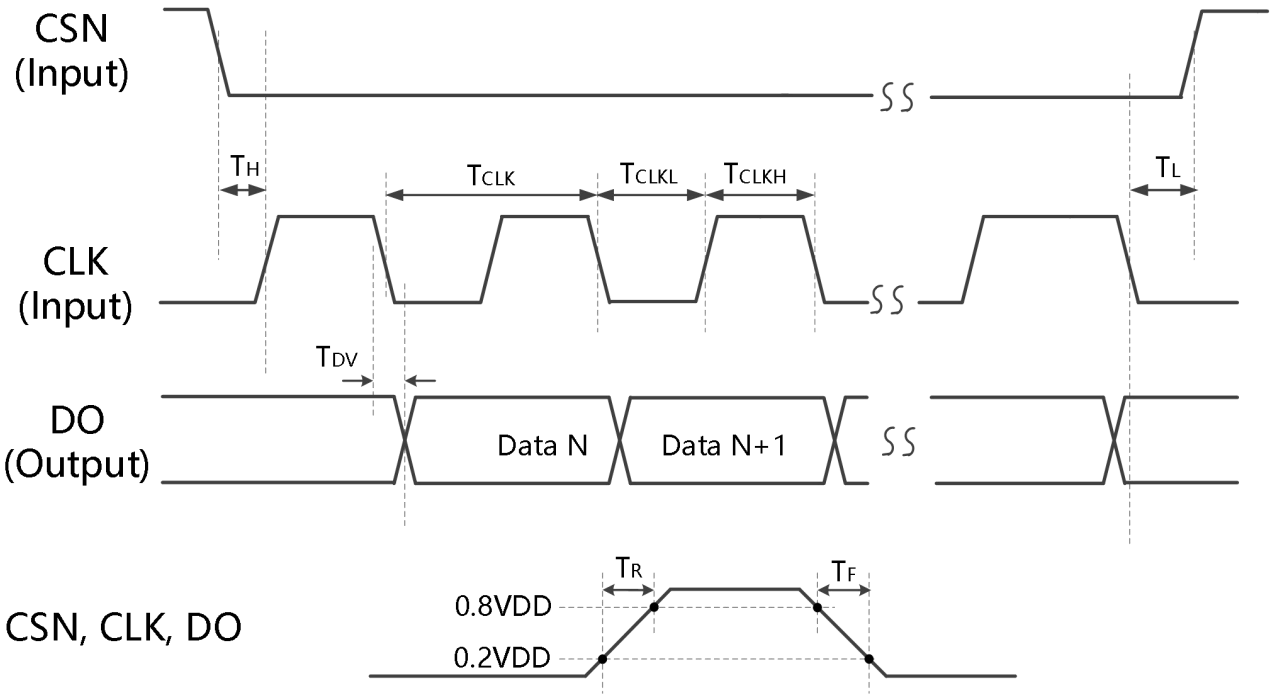



***Figure-24: SSI Timing Diagram***

***SSI Timing Parameter***

| Symbol | Notes | Min. | Typ. | Max. | Unit |
|---|---|---|---|---|---|
| $T_H$ | Time between CSN falling edge and CLK Rising edge | 100 | | - | ns |
| $T_{CLK}$ | Clock period | 64 | | - | ns |
| $T_{CLKL}$ | Low period of clock | 30 | | - | ns |
| $T_{CLKH}$ | High period of clock | 30 | | - | ns |
| $T_L$ | Time between CLK last falling edge and CSN rising edge | $0.5 \cdot T_{CLK}$ | | - | ns |
| $T_R$ | Rise Time of Digital Signal (with 20pf Loading Condition) | - | 10 | - | ns |
| $T_F$ | Fall Time of Digital Signal (with 20pf Loading Condition) | - | 10 | - | ns |
| $T_{DV}$ | Data valid time of DO (with 20pf Loading Condition) | - | - | 15 | ns |

### 7.8.2 SSI Read Angle

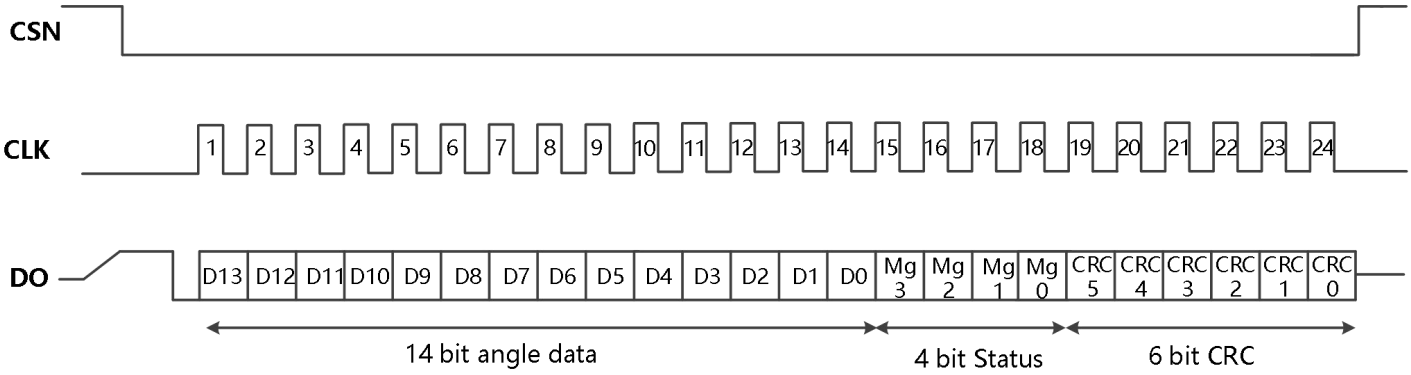



***Figure-25: SSI Data Format***

An SSI data transfer starts when CSN is pulled to logic 'Low', as shown in Figure-25. CLK is the Serial Port Clock, and it is controlled by the SSI master; it is high when there is no SSI transmission. DO (Data Output) is the Serial Port Data Output from MT6701, it is driven at the rising edge of CLK and should be captured at the falling edge of CLK.

**Bit 0-13**: 14-bit Angle Data D[13:0]
**Bit 14-17**: 4-bit Magnetic Field Status Mg[3:0]
**Bit 18-23**: 6-bit CRC Code CRC[5:0]

0~360° absolute angle θ could be calculated by the below formula with D[13:0]:

$$\theta = \frac{\sum_{i=0}^{13} D<i> \bullet 2^{i}}{16384} \bullet 360°$$

Mg[3:0] Magnetic Field Status Truth Table:

| Mg[1:0] | Status |
|---|---|
| 0 | Normal |
| 1 | Magnetic Field is too Strong |
| 2 | Magnetic Field is too Weak |
| 3 | - |

| Mg[2] | Status |
|---|---|
| 0 | Normal |
| 1 | Push Button is Detected |

| Mg[3] | Status |
|---|---|
| 0 | Normal |
| 1 | Loss of Track |

CRC Data Range: D[13:0] and Mg[3:0] total 18-bit, D[13] is the MSB, Mg[0] is the LSB
CRC polynomial: $X^6+X+1$, MSB steam in first.

## 7.9 Pushbutton Output

The MT6701 implements a pushbutton detection function through a dynamic and relative measurement of the orthogonal magnetic field strength. This pushbutton detection function drives the PUSH output pin high when the MT6701 detects a fast (which is less than the time defined by 'PUSH_DIFF_DLY') increase of the magnetic field (which is larger than the percentage defined by 'PUSH_THRD'). After a fast decrease of the magnetic field, the PUSH output is driven low.

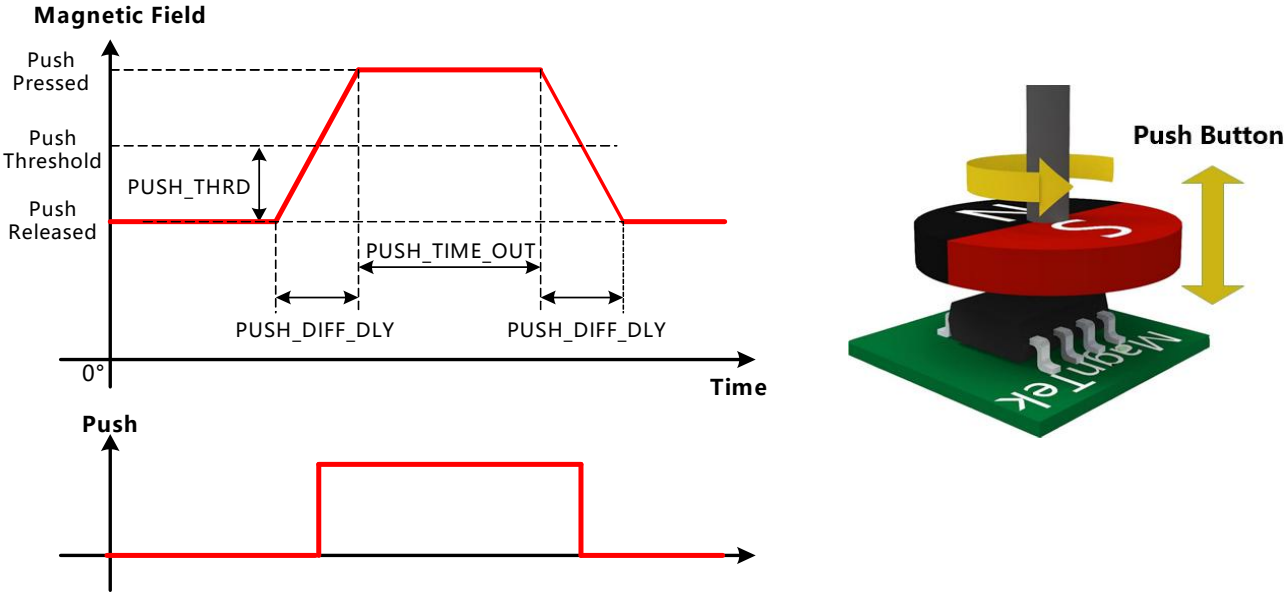



*Figure-26: Pushbutton Function*

***PUSH_THRD Register (EEPROM)***

| Reg. PUSH_THRD | Change Percentage |
|---|---|
| 0 | +44% |
| 1 | +50% |
| 2 | +38% |
| 3 | +31% |

***PUSH_DIFF_DLY Register (EEPROM)***

| Reg. PUSH_DIFF_DLY | Time (Second) |
|---|---|
| 0 | 0.5 |
| 1 | 0.25 |

***PUSH_TIME_OUT Register (EEPROM)***

| Reg. PUSH_TIME_OUT | Time (Second) |
|---|---|
| 0 | 8 |
| 1 | 16 |
| 2 | 4 |
| 3 | 2 |

# 8. Register Map & EEPROM Programming

## 8.1 EEPROM Register Map

| Reg. Addr. | bit7 | bit6 | bit5 | bit4 | bit3 | bit2 | bit1 | bit0 |
|---|---|---|---|---|---|---|---|---|
| 0x25 | UVW_MUX | | | | | | | |
| 0x29 | | ABZ_MUX | | | | | DIR | |
| 0x30 | UVW_RES[3:0] | | | | | | ABZ_RES[9:8] | |
| 0x31 | ABZ_RES[7:0] | | | | | | | |
| 0x32 | HYST[2] | Z_PULSE_WIDTH[2:0] | | | ZERO [11:8] | | | |
| 0x33 | ZERO [7:0] | | | | | | | |
| 0x34 | HYST[1:0] | | | | | | | |
| 0x38 | PWM_FREQ | PWM_POL | OUT_MODE | | | | | |
| 0x3E | A_STOP[11:8] | | | | A_START[11:8] | | | |
| 0x3F | A_START[7:0] | | | | | | | |
| 0x40 | A_STOP[7:0] | | | | | | | |

***Warning: the register bits which filled by ■ are reserved by MagnTek, please DO NOT change the value of these bits!!!***

### *UVW_MUX (Address 0x25[7])

UVW_MUX register contains the configuration data of UVW output type.

| UVW_MUX | UVW Output Type(Only for QFN Package) |
|---|---|
| 0x0 | UVW |
| 0x1 | -A-B-Z |

### *ABZ_MUX (Address 0x29[6])

ABZ_MUX register contains data of ABZ output type.

| ABZ_MUX | ABZ Output Type |
|---|---|
| 0x0 | ABZ |
| 0x1 | UVW |

### *DIR(Address 0x29[1])

DIR register contains the configuration data of output rotation direction

| DIR | Output Direction |
|---|---|
| 0x0 | CCW |
| 0x1 | CW |

### *UVW_RES[3:0] (Address 0x30[7:4])

UVW_RES register contains the configuration data of UVW output resolution (Pole-Paris).

| Reg. UVW_RES<3:0> | UVW Output Pole Pairs |
|---|---|
| 0x0 | 1 |
| 0x1 | 2 |
| 0x2 | 3 |
| ..... | ..... |
| 0xD | 14 |
| 0xE | 15 |
| 0xF | 16 |

### *ABZ_RES[9:0] (Address 0x30[1:0] & 0x31[7:0])

ABZ_RES register contains the configuration data of ABZ output resolution (PPR).

| Reg. ABZ_RES<9:0> | ABZ Resolution (Pulse per Round) |
|---|---|
| 0x000 | 1 |
| 0x001 | 2 |
| 0x002 | 3 |
| ..... | ..... |
| 0x3FD | 1022 |
| 0x3FE | 1023 |
| 0x3FF | 1024 |

### *HYST[2:0] (Address 0x32[7] & 0x34[7:6])

HYST register contains the configuration data of hysteresis filter parameter.

| HYST | Hysteresis (LSB) |
|---|---|
| 0x0 | 1 |
| 0x1 | 2 |
| 0x2 | 4 |
| 0x3 | 8 |
| 0x4 | 0 |
| 0x5 | 0.25 |
| 0x6 | 0.5 |
| 0x7 | 1 |

### *Z_PULSE_WIDTH[2:0] (Address 0x32[6:4])

Z_PULSE_WIDTH register contains the configuration data of Z pulse width (Fig.9 & Fig.10 )

| HYST | Z Pulse Width |
|---|---|
| 0x0 | 1 LSB |
| 0x1 | 2 LSB |
| 0x2 | 4 LSB |
| 0x3 | 8 LSB |
| 0x4 | 12 LSB |
| 0x5 | 16 LSB |
| 0x6 | 180° |
| 0x7 | 1 LSB |

### *ZERO[11:0] (Address 0x32[3:0] & 0x33[7:0])

ZERO register contains the configuration data of zero-degree position.

| ZERO | Zero Degree Position |
|---|---|
| 0x000 | 0° |
| 0x001 | 0.088° |
| 0x002 | 0.176° |
| ... | ... |
| 0xFFD | 359.736° |
| 0xFFE | 359.824° |
| 0xFFF | 359.912° |

### *PWM_FREQ (Address 0x38[7])

PWM_FREQ register contains the configuration data of PWM frame frequency

| PWM_FREQ | PWM Frame Frequency |
|---|---|
| 0x0 | 994.4 Hz |
| 0x1 | 497.2 Hz |

### *PWM_POL (Address 0x38[6])

PWM_POL register contains data of PWM polarity.

| PWM_POL | PWM Polarity |
|---|---|
| 0x0 | High Level Valid |
| 0x1 | Low Level Valid |

### *OUT_MODE(Address 0x38[5])

OUT_MODE register contains the configuration data of 'Out' Pin Mode

| OUT_MODE | 'Out' Pin Mode |
|---|---|
| 0x0 | Analog Output |
| 0x1 | PWM Output |

### *A_START[11:0] (Address 0x3E[3:0] & 0x3F[7:0])

A_START register contains the configuration data of start-point of analog output (Fig.15)

| A_START | Analog/PWM Start Angle |
|---|---|
| 0x000 | 0° |
| 0x001 | 0.088° |
| 0x002 | 0.176° |
| ... | ... |
| 0xFFD | 359.736° |
| 0xFFE | 359.824° |
| 0xFFF | 359.912° |

### *A_STOP[11:0] (Address 0x3E[7:4] & 0x40[7:0])

A_STOP register contains the configuration data of stop-point of analog output (Fig.15)

| A_STOP | Analog/PWM Stop Angle |
|---|---|
| 0x000 | 0° |
| 0x001 | 0.088° |
| 0x002 | 0.176° |
| ... | ... |
| 0xFFD | 359.736° |
| 0xFFE | 359.824° |
| 0xFFF | 359.912° |

## 8.2 EEPROM Programming

| Reg. Addr. | bit7 | bit6 | bit5 | bit4 | bit3 | bit2 | bit1 | bit0 |
|---|---|---|---|---|---|---|---|---|
| 0x25 | UVW_MUX | | | | | | | |
| 0x29 | | ABZ_MUX | | | | | DIR | |
| 0x30 | UVW_RES[3:0] | | | | | | ABZ_RES[9:8] | |
| 0x31 | ABZ_RES[7:0] | | | | | | | |
| 0x32 | HYST[2] | Z_PULSE_WIDTH[2:0] | | | ZERO [11:8] | | | |
| 0x33 | ZERO [7:0] | | | | | | | |
| 0x34 | HYST[1:0] | | | | | | | |
| 0x38 | PWM_FREQ | PWM_POL | OUT_MODE | | | | | |
| 0x3E | A_STOP[11:8] | | | | A_START[11:8] | | | |
| 0x3F | A_START[7:0] | | | | | | | |
| 0x40 | A_STOP[7:0] | | | | | | | |

***Warning: the register bits which filled by ■ are reserved by MagnTek, please DO NOT change the value of these bits!!!***

Customer can program the EEPROM registers by I²C interface (Fig.17 & Fig.18) and follow the below steps. **When programming EEPROM, must keep 4.5V<VDD<5.5V**.

| Step | Operation |
|---|---|
| **Writing Register** | Writing the right value to the target EEPROM registers (Do Not Change the ■ bits, customer should read out the ■ bits and logic these bits 'or' with others) |
| **Programming Key** | Write value '0xB3' to Register '0x09' |
| **Programming Command** | Write value '0x05' to Register '0x0A' |
| **Programming (VDD>4.5V)** | Waiting >600ms without any operation to MT6701 |
| **Check Programming Data** | Power-down MT6701 and then power-up it again, read the EEPROM register to check if the data is successfully programmed |

# 9. Mechanical Angle Direction

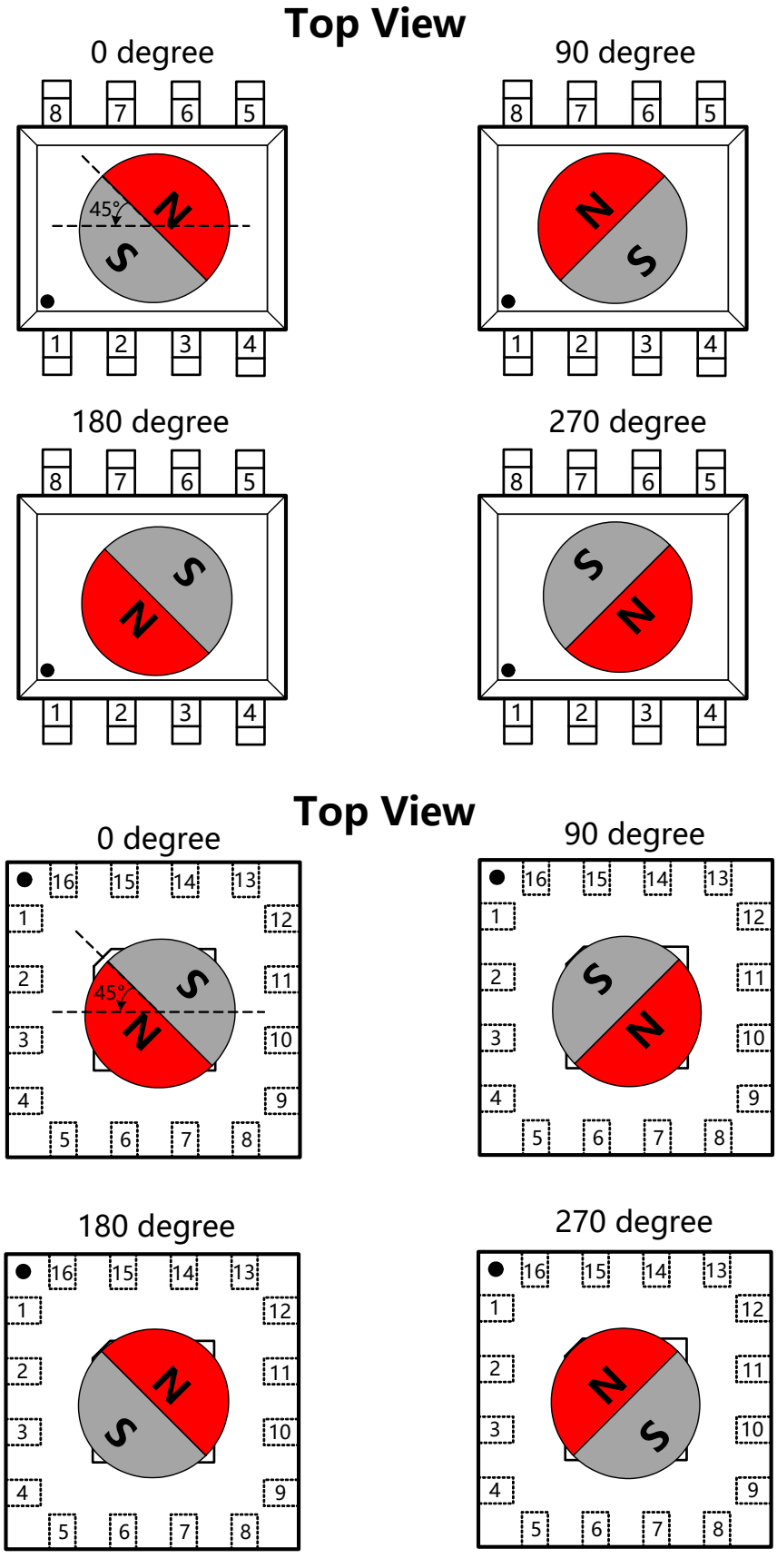
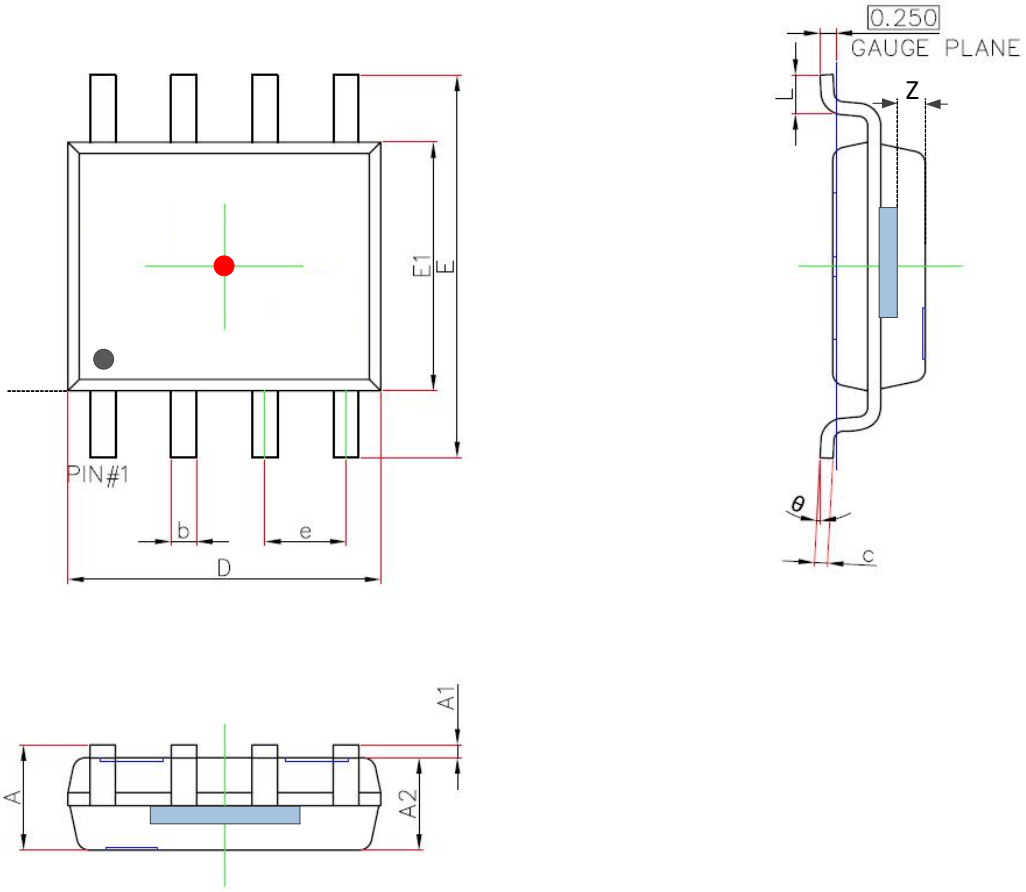



*Figure-27: Mechanical Angle Position*

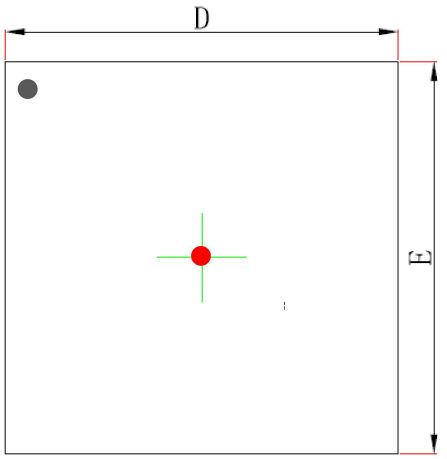
# 10. Package Information

## 10.1 SOP-8 Package

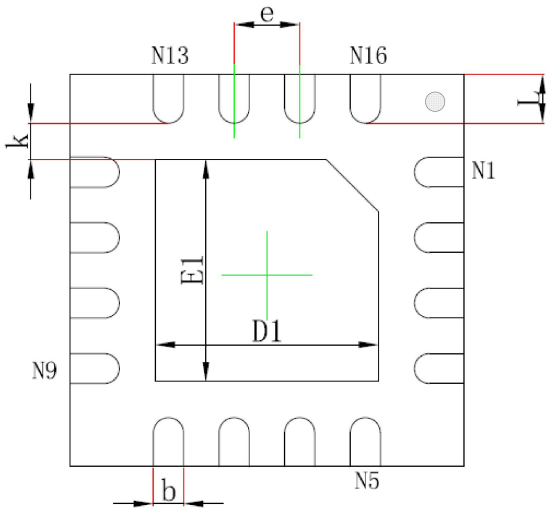



| Symbol | Dimensions in Millimeters | | Dimensions in Inches | |
|---|---|---|---|---|
| | Min. | Max. | Min. | Max |
| A | 1.450 | 1.750 | 0.057 | 0.069 |
| A1 | 0.100 | 0.250 | 0.004 | 0.010 |
| A2 | 1.350 | 1.550 | 0.053 | 0.061 |
| b | 0.330 | 0.510 | 0.013 | 0.020 |
| c | 0.170 | 0.250 | 0.007 | 0.010 |
| D | 4.700 | 5.100 | 0.185 | 0.201 |
| E | 5.800 | 6.200 | 0.228 | 0.244 |
| E1 | 3.800 | 4.000 | 0.150 | 0.157 |
| e | 1.270(BSC) | | 0.050(BSC) | |
| L | 0.400 | 1.270 | 0.016 | 0.050 |
| θ | 0° | 8° | 0° | 8° |
| Z | 0.45 | 0.65 | 0.018 | 0.026 |

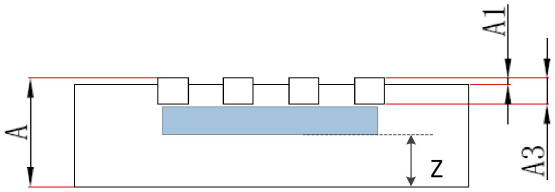
### 10.2 QFN-16 Package




Top View




Bottom View




| Symbol | Dimensions in Millimeters | | Dimensions in Inches | |
|---|---|---|---|---|
| | Min. | Max. | Min. | Max. |
| A | 0.700 | 0.800 | 0.028 | 0.031 |
| A1 | 0.000 | 0.050 | 0.000 | 0.002 |
| A3 | 0.203REF | | 0.008REF | |
| D | 2.900 | 3.100 | 0.114 | 0.122 |
| E | 2.900 | 3.100 | 0.114 | 0.122 |
| D1 | 1.600 | 1.800 | 0.063 | 0.071 |
| E1 | 1.600 | 1.800 | 0.063 | 0.071 |
| k | 0.275REF | | 0.011REF | |
| b | 0.180 | 0.300 | 0.007 | 0.012 |
| e | 0.500REF | | 0.020REF | |
| L | 0.300 | 0.500 | 0.012 | 0.020 |
| Z | 0.220 | 0.420 | 0.009 | 0.017 |

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# 12. Revision History

| Revision Number | Date | Comments |
|---|---|---|
| 1.0 | 2020.03 | Initial Release |
| 1.1 | 2020.06 | Update EEPROM Programming Condition :<br>4.5V<VDD<5.5V |
| 1.2 | 2020.07 | Update POD |
| 1.3 | 2020.09 | Update Mechanical Angle |
| 1.4 | 2021.01 | Update PWM Description |
| 1.5 | 2021.03 | Update PWM Clock Duration |
| 1.6 | 2022.05 | Add Part Number List<br>Add Power-on AB Pulses Indicating Absolute Position |
| 1.7 | 2022.07 | Update SSI Data Format, Figure-25 |
| 1.8 | 2022.12 | Update SSI Timing, Figure-24 |